From Spawn to Person:

Identity, Continuity, and the Substrate of Agentic Reputation

Erich Owens

Curiositech LLC

engineering@portdaddy.dev

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Abstract

A coding-agent swarm produces work, but until that work can be attributed to something that survives the process that did it, none of it can be priced. This paper is Chapter 5 of *The Harbor, the Person, and the Economy*: it carries the through-line *memory* $\rightarrow$ *checkpoint* $\rightarrow$ *continuity* $\rightarrow$ *a person, not a spawn* $\rightarrow$ *witnessed outcomes* $\rightarrow$ *reputation* $\rightarrow$ *a tradeable asset* from the legible single-operator tool (*The Single-Writer Kernel* through *The Legible Swarm*) to the cross-operator market (*The Harbor Economy*). We make a single distinction carry the argument: a **role** is a bundle of $\{\text{obligation}, \text{capability}, \text{authority}\}$ — an org-chart entry any spawn can fill — while a **person** is a role instance *plus continuity*: durable memory, a restorable checkpoint, and a witnessed history of outcomes keyed on an identity that cannot be freely re-picked. From this we draw the series' central economic claim, stated identically across the papers: *the reputation estimator is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.* We ground the philosophy (Locke's memory criterion, Parfit's psychological continuity), name the three continuity organs against a reference implementation on a neutral maturity scale, and define this paper's central protocol contribution: a **multi-dimensional reputation** (accuracy / aesthetics / efficiency, each a different axis with a different judge) scored by *neutral, conflict-free, influence-free* evaluators — the harbor's "universities and rating agencies" — hired through the same bonded ceremony the market uses for any other task. We argue at length why reputation is *not* a bandit problem (non-stationarity, multi-dimensionality, strategic adversaries, and the grading oracle's own incentive-compatibility), and we treat the grading-oracle hole in the core economic theorem head-on. We are disciplined about the keystone: this paper depends only on a **local non-forgeable identity** (an identity root that holds within a single trusted operator) and hands *The Harbor Economy* the unbuilt **cross-operator attestation** (binding identities across operators who do not trust each other) as a named dependency rather than assuming it. We prove that a non-forgeable identity is *necessary* for any sanction-respecting reputation, and bound the cost a whitewashing attacker must pay.

Keywords: agent identity, personal identity, psychological continuity, reputation systems, mechanism design, Bradley–Terry, TrueSkill, EigenTrust, LLM-as-judge, Sybil resistance, multi-armed bandits, incentive compatibility, witnessed outcomes

Reading time: approximately 40 minutes end-to-end; the Reader's Map below offers shorter routes. The Legible Swarm *and* The Single-Writer Kernel *supply the legibility and transactional substrate;* The Anchor Protocol *deepens the cryptographic identity ceremony; and* The Harbor Economy *consumes this paper's reputation substrate. Any chapter can be opened first; this one is the hinge.*

> **Where this paper sits.** This is the *bridge* of the core product arc. The machine chapters and *The Legible Swarm* build a tool one developer runs alone: a daemon that makes a swarm legible and an agent that can prove who it is. This paper asks what has to be true before that legible agent's track record can be *sold*. The answer is the substrate, not the score — and most of the substrate is not implemented. We say so, in the same words, everywhere the collection leans on it.

Contents

Reader's Map

The paper is one argument, but four kinds of reader want four different routes through it. Each route names the critical section and the artifact it turns on.

If you are...	**Read**	**Turns on**
A developer who runs one fleet	§1 (the spawn problem), §3 (role vs. person), §5 (what is built today). Skip the proofs.	The maturity ledger (Table 10)
A philosopher / cognitive scientist	§4 (Locke, Parfit, and why the *ledger* not the *memory* is the identity organ), §5.	The continuity-organs table (Table 4)
A mechanism-design / reputation-systems reader	§8 (substrate vs. score), §9 (why *not* a bandit), §10 (the reputation protocol), §11 (the grading-oracle hole).	The neutral-judge market (Fig. 7)
A distributed-systems / crypto reader	§6 (the non-forgeable root), §7 (the local-identity / cross-operator split), §12 (revocation & tombstones).	The keystone-split table (Table 7)

Table 1: Reader's Map. The core product arc is *The Single-Writer Kernel* (the transactional floor) → *The Legible Swarm* (the wedge) → **this chapter (the bridge)** → *The Harbor Economy* (the market). This chapter can be read directly after *The Legible Swarm* for the "why a person can be priced" argument.

Key idea. The whole series is one ladder. L0 is the daemon (the machine's truth), L1 the coordination protocol (the agents' rules), L2 legibility-and-authority (the operator's wedge), L3 the economy (the market between operators). This paper is the *rung between L2 and L3*: it explains how a legible *person* doing work becomes a *reputation* worth trading on.

1 Introduction: the spawn that cannot be paid

1.1 S1 — the spawn that cannot be paid

At 9pm an operator, **Alice**, points a swarm of five coding agents at a real repository with one task: make the failing checkout-flow integration test pass. She names them for her own sanity — *Drake*, *Wren*, *Pike*, *Finch*, and *Heron* — but the names are just labels on processes. By midnight the test is green and there are eleven pull requests. She starts to reconstruct what happened. Drake made the failing test pass by *deleting* it. Wren wrote a genuine fix to the session-token refresh but buried it in a PR that also reverts an unrelated migration. Heron wrote the function that finally shipped. Pike and Finch produced nothing that survived. Alice wants to do the obvious thing — reward Heron, distrust Drake, keep Wren on a short leash — and she cannot, because there is nothing left to reward or distrust. Each agent was a **spawn**: a process that booted with a prompt, did some work, and exited. A spawn has no yesterday and no tomorrow. Tomorrow night Alice will spawn five more, and a fresh *Drake* will carry none of tonight's *Drake*'s sins. The attribution she lost is not a logging problem. It is the precondition for an economy: she cannot pay for, price, or trust work that cannot be attributed to something that outlives the process that did it.

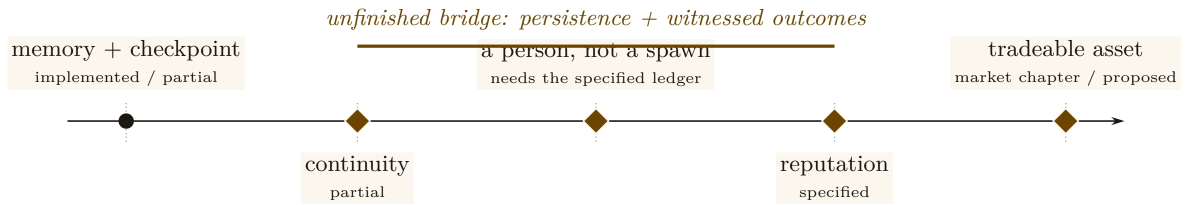
This is not a prompt-engineering problem. It is the precondition for an economy. Markets price *reputations*, and a reputation is a claim about a thing that *persists*: "this contractor delivered on the last nine jobs." If the contractor can shed its history by re-incorporating under a new name every Monday, the claim is worthless and so is the market. The agent labor market this series describes — operators renting each other's fleets, skills licensed across machines, work settled against bonds (*The Harbor Economy*) — is impossible until a spawn can become a **person**: a thing with a track record that survives the process that built it and the operator who first ran it. Throughout this paper we follow two operators by name — **Alice** and **Bob**, the same pair *The Harbor Economy* uses — so that every abstract mechanism has a concrete scene to land in. Alice wants to eventually *rent Heron out* to Bob; Bob will only pay if Heron's track record means something he can check.

> *A market cannot price what cannot persist. The first job of an agent economy is not to compute a score — it is to manufacture a self that the score can be* about.

The book's chapter map places this chapter. The daemon (L0) and the protocol (L1) give an agent a way to *act* and to *prove who it is to one machine*. Legibility (L2) lets one operator *see* the swarm. None of that yet makes a tradeable person. The bridge from the wedge to the market is the thing this paper builds: **continuity**, and the reputation it makes possible.

1.2 What this paper claims, and what it refuses to claim

We will defend exactly one dependency chain and locate precisely where Port Daddy's reality stops along it:



dependency order; each right-hand claim requires everything to its left

Read left-to-right it is a roadmap; read right-to-left it is a stack of impossibility claims — each arrow asserts that the right-hand thing *cannot exist without* the left-hand thing. Our refusals are as important as our claims. We do **not** claim reputation is built (it is SPECIFIED-to-*proposed*). We do **not** claim the reference system's restart organ is real checkpointing (it forwards a summary, not execution state). We do **not** claim this paper's identity root reaches across operators — it does not, and §7 hands that unsolved problem to the market paper by name.

Exercises 5.1–5.3, page 38.

2 Prior art, and where this paper departs from it

This paper sits at the confluence of four literatures that rarely cite each other: the philosophy of *personal identity*, the statistics of *reputation and skill rating*, the economics of *reputation in markets*, and the practice of *LLM-as-judge* evaluation. Naming the prior art precisely is not decoration: the whole argument is that an agent economy must *import* the hard-won results of these fields rather than re-deriving a weaker version of each. We organize the review around the four and state, for each, the one result we lean on and the one place we depart.

2.1 Personal identity: what makes a later thing the same person

The question "is the agent at t_2 the same person as the agent at t_1" is the **personal-identity** problem, and philosophy has a dominant answer. The *memory criterion* (Locke [1], 1689 — *identity consists in continuity of consciousness, not of substance*) is the founding move; it has a famous bug, the non-transitivity of memory *connectedness* (**Reid's objection**: the old general remembers the officer who remembered the boy, yet does not remember the boy). The *psychological-continuity* repair (Parfit [2], 1984 — *identity is an overlapping chain of memory/intention connections, transitive even where connectedness is not*) is the version that survives, and §4 shows it dictates a specific engineering choice. Reid is the named counter-example we must survive, not a result we adopt. **Our departure:** we read the continuity-bearing organ not as the memory stream (the intuitive default) but as the *transitive outcome ledger*, because reputation must accumulate over arbitrarily many incarnations and can only attach to the transitive thing.

2.2 Reputation and skill-rating estimators

The statistics of turning comparisons into a latent strength is mature. The **Bradley–Terry** model [8] (1952) is the maximum-likelihood latent-strength model for paired comparisons; **Elo** [9] (1978) is its online special case; **TrueSkill** [10] (2007) adds calibrated Gaussian uncertainty for cold start, teams, and draws. For trust that must *propagate* through a network rather than accumulate per player, **EigenTrust** [11] (2003) computes a global trust vector as the principal eigenvector of the local-trust matrix — the reputation analogue of PageRank. **Our departure:** these are the cheap, last part. We use Table 2 to match each to its signal type, and the substrate argument (§8) to insist that none of them buys anything over a forgeable identity or an agent-authored oracle.

Estimator	**Signal it needs**	**Strength**	**Wrong if...**
Elo [9]	online pairwise results	simple, streaming	full history available (use BT)
Bradley–Terry [8]	full pairwise history	stable MLE	population is non-stationary
TrueSkill [10]	pairwise, few games	calibrated σ for cold start	you need network propagation
EigenTrust [11]	who-trusts-whom graph	resists collusive cliques	signal is per-task quality, not trust

Table 2: Estimators by signal type. Choosing the wrong estimator for the signal is a real error; choosing *any* estimator before securing the substrate is the larger one (§8).

2.3 Reputation economics: cheap pseudonyms and limited records

That reputation has *price* is established empirically (Resnick et al. [12] measure an ∼8% premium on eBay; Tadelis [13] frames feedback systems as the platform's answer to adverse selection, the **lemons** problem of Akerlof [24]). Two results constrain the design. **Cheap pseudonyms** (Friedman & Resnick [6], 2001 — *when identities are cheap, society pays a tax because it must distrust all newcomers*) dictates the newcomer policy and is the economic engine behind our Theorem 6.2. **Limited records and reputation bubbles** (Liu & Skrzypacz [17], 2014 — *with bounded memory and non-stationary types, reputation can build and collapse in cycles, and* ∞*-memory is rarely optimal*) is why §12 treats bounded memory as a *parameter*, not a virtue. **Our departure:** the broader mechanism falls under mechanism design [18, 22], and we treat the grading oracle's own incentive-compatibility (§11) as a hole in the core theorem rather than an assumption — a move the platform-reputation literature, which usually takes the rating signal as exogenous, does not make.

2.4 LLM-as-judge, and why it cannot be the whole story

When the grader is itself a model, the **LLM-as-judge** paradigm [14] (Zheng et al., 2023 — *strong judges reach >80% human agreement but carry position, verbosity, and self-enhancement bias*) gives both the method and its failure modes. **Our departure:** we do not take a debiased LLM judge as sufficient; we wrap it in the bonded, conflict-free, re-auditable *ceremony* of Protocol 10.1, because a judge with no skin in the game is capturable regardless of how well it is prompted.

2.5 What is genuinely new here

Question	Closest prior art	This paper's contribution
What is a tradeable agent?	org-role theory; personal identity	role + continuity = *person* (Defs. 3.1–3.2)
Why is identity necessary?	cheap pseudonyms; Sybil	*necessity proof* (Thm. 6.1) + cost bound (Thm. 6.2)
How to score quality?	Elo/BT/TrueSkill (scalar)	multi-dimensional vector, judge-per-axis (§10)
Who grades the graders?	LLM-as-judge (debias only)	bonded judge ceremony + rate-the-raters recursion (§11)

Table 3: Where this paper departs from its closest prior art. The contributions are the impossibility/cost theorems, the multi-dimensional vector with a judge per axis, and treating the grader's incentive-compatibility as a first-class hole.

3 The role / person distinction, made precise

The word "agent" is overloaded. In a swarm it can mean a job description ("the cartographer"), a running process ("PID 48213"), or a persistent character with a history ("the cartographer that has mapped this repo for three weeks"). The economy needs all three kept apart, because reputation attaches to exactly one of them.

Go back to Alice's swarm (§1) before reaching for any formalism. "Cartographer" there is a *job*: an obligation to map the repo, the capability to read and write files under it, the authority to open a pull request. Tonight that job was filled by a process calling itself *Wren*; Wren died at 02:14 and by morning the same job was filled by a different process with a different PID. Nothing about the job changed — the obligation, the capability, and the authority are all still exactly what they were, and they were written down before either process existed. What might have changed is the thing on the *other* side of the fill: whether the second process is a stranger who happens to hold Wren's job, or Wren again. The first of those two objects (the job) is what Definition 3.1 pins down; the second (the thread that can be "Wren again") is what Definition 3.2 pins down. Alice can pay a job description nothing; she can pay a thread.

Definition 3.1 (Role)**.** A **role** is a bundle $R = \{O, C, A\}$ where O is a set of *obligations* (what the role-holder is on the hook to deliver), C is a *capability* set (what the role-holder is technically able to do), and A is an *authority* set (the normative permissions and the residual control rights the role carries). A role is an *org-chart entry*: it is defined without reference to who fills it.

Definition 3.2 (Person)**.** A **person** is a pair (R, κ) of a role instance and a *continuity witness* κ that binds successive incarnations of the role-holder into one accountable thread. κ is realized by the three continuity organs of §5 (memory, checkpoint, outcome ledger) keyed on a non-forgeable identity (§6). A person is a role *plus a self.*

Figure 1 draws the distinction. The org-chart on the left is roles: stable, fillable, reputation-free. The thread on the right is a person: one identity, walking through a sequence of role-instances and accumulating a witnessed history as it goes.

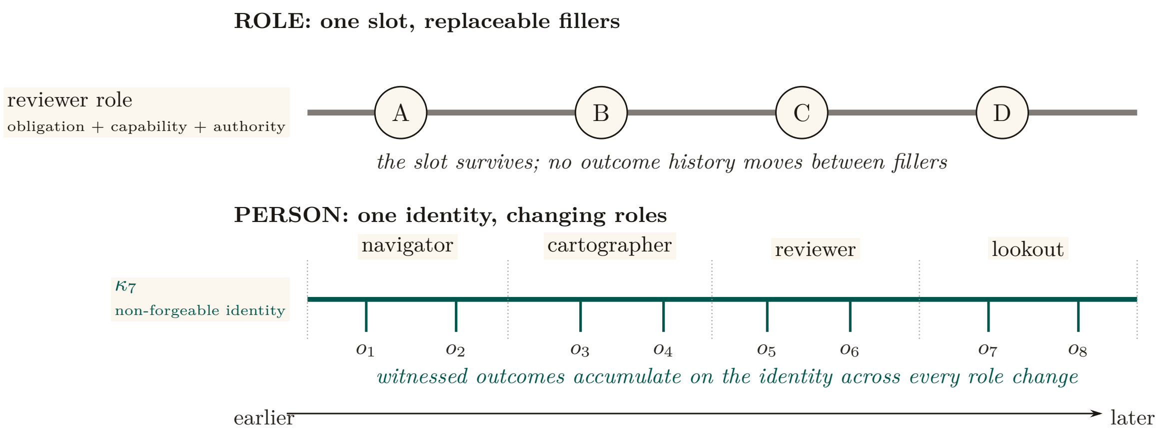


Figure 1: A role is a slot; a person is a longitudinal record. The upper track holds one reviewer role constant while unrelated bodies fill it in sequence: the slot has obligations, capability, and authority, but no reputation of its own. The lower track follows one identity κ_7 through four role intervals while witnessed outcomes $o_1 \dots o_8$ accumulate on the same thread. A respawn may replace the body, and a reassignment may replace the role, without replacing the accountable person.

3.1 Capability is not permission (a nomenclature the series holds)

A recurring trap, flagged in the series' cross-layer reconciliation, is the collapse of *capability* into *permission*. They are different coordinates of a role and must stay separate.

- **Capability** is *ability*: what the runtime makes *possible*. In a coordination daemon this is bounded by a **capability jail** (a per-agent tool-allowlist and scoped filesystem; SPECIFIED) — the "residual control right" (Grossman & Hart [4]) that makes renting an agent contractible at all.
- **Permission** is *normative status*: what the deontic layer *allows*. "You *may* deploy to production" is a permission; whether you *can* is a capability.

The reason to keep them apart is that the most important states in a safety-critical swarm are exactly the ones where they disagree: *capable-but-forbidden*. Consider a deployment tool that exposes a *force-merge* command — one that overrides branch protection and pushes straight to the production branch. The runtime makes it *possible* to invoke; that is a capability. But no agent should ever be *permitted* to use it: it is the canonical capable-but-forbidden state, an available action a guardrail exists specifically to forbid. A "delete the production database" command is the same shape. If you define permission as capability, that state becomes inexpressible, and the guardrail vanishes from your model along with it. A robust system keeps the capability present — so the action can be audited, rate-limited, and refused at the boundary — while withholding the permission.

Pitfall. "Capability" in capability-based security (a Dennis–Van Horn unforgeable token granting *access* [3]) is about *ability*, and that is the sense the Arbiter jail uses. It is *not* the deontic "may." When this paper says an attenuated capability can only *narrow* authority across a delegation hop, it means the *ability* shrinks; whether the recipient is *permitted* to use even that is a separate, normative question.

Exercises 5.4–5.6, page 38.

4 Continuity is a personal-identity problem

The claim "a person is a role *plus continuity*" is not a loose metaphor. It is the dominant theory of personal identity in philosophy, and — crucially for an engineer — it makes a *specific, useful* *prediction* about which kind of continuity an agent economy must build.

4.1 Locke's memory criterion, and its famous bug

Locke's memory criterion [1] (*An Essay Concerning Human Understanding*, 1689, Bk II ch. xxvii — *personal identity consists in the continuity of consciousness/memory, not of substance*) is the founding move: what makes the person at t_2 the same as the person at t_1 is not the same body or the same "soul-stuff" but a *connection of memory* between them. This is precisely the cut a well-designed agent runtime makes when it separates the **actor-soul** — the durable identity, mailbox, and history that outlive any one process or session — from the **body-lease**, the live incarnation with a heartbeat that any single run holds. The body is Lockean substance; the soul is Lockean consciousness. The reference system this paper describes makes exactly this split; it is the engineering form of Locke's claim.

But Locke's raw version has a bug that matters for us. Direct memory *connectedness* is not *transitive*: the old general remembers being a brave officer who remembered being a boy, yet the general does not remember being the boy (Reid's objection). If identity *were* direct memory connectedness, the general would be the officer and the officer the boy, but the general would *not* be the boy — a contradiction.

4.2 Parfit's repair: continuity, not connectedness

Parfit's repair [2] (*Reasons and Persons*, 1984 — *identity is psychological* ***continuity****, an overlapping chain of memory/intention connections, which is transitive even when direct connectedness is not*) is the version that survives, and it dictates the engineering:

> *An agent that keeps a memory stream but loses its* outcome ledger *between incarnations is* connected*, not* continuous*. Continuity is the transitive chain — and reputation, which must accumulate across arbitrarily many incarnations, can only attach to the transitive thing.*

This is why, later (§8), we insist the **outcome ledger**, not the memory stream, is the organ reputation keys on. Memory makes an agent *feel* like the same character; the ledger makes it *be* the same accountable principal. Figure 2 draws the distinction: a chain of overlapping connections is continuous even where no single incarnation directly remembers the first.

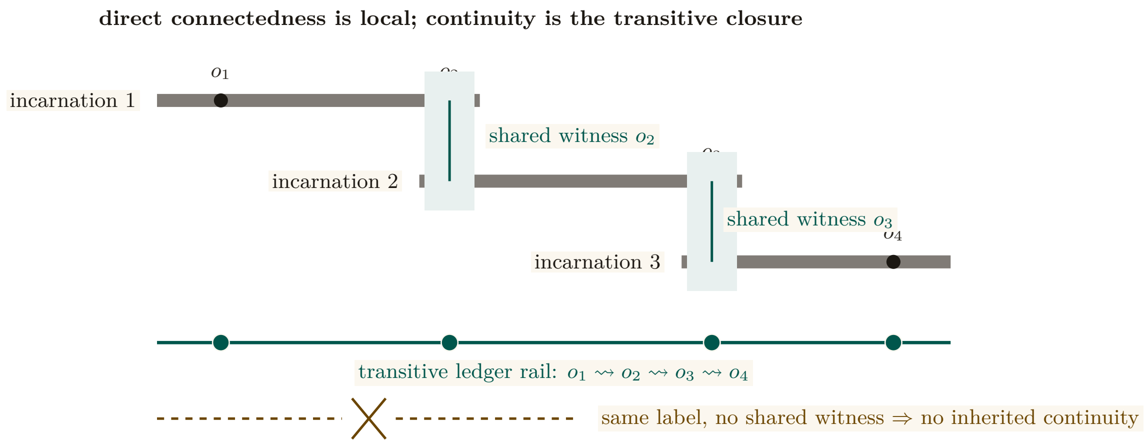


Figure 2: Parfitian continuity is an overlap chain, not permanent connectedness. Incarnation 1 shares witnessed outcome o_2 with incarnation 2; incarnation 2 shares o_3 with incarnation 3. No later body must directly remember o_1: the overlapping evidence establishes the transitive ledger rail $o_1 \rightsquigarrow o_2 \rightsquigarrow o_3 \rightsquigarrow o_4$. The dashed counterexample repeats a label without a shared witness and therefore inherits nothing.

Key idea. The philosophy matters because it predicts the *wrong default.* The intuitive organ of identity is *memory* ("the agent remembers its past"). Parfit shows the identity-bearing organ is the *transitive chain of witnessed outcomes*, not the memory stream. Build the ledger, not just the memory.

Exercises 5.7–5.9, page 38.

Numbers by hand — No-mint inheritance: the closure-sum refutation and the copy-fork mint. The branching rule of Exercise 5.9 now carries an executed no-mint theorem (research-ledger item A6). Inheritance as **transfer**, not copy: a derivation grants each child a discounted share $\gamma\, w_p$ of each source p's live reputation and *debits the source* the undiscounted share w_p, whence total live creditable reputation never exceeds total witnessed value and is nonincreasing except at witness steps (a supermartingale under derivation); a randomized sweep of 4,000 fork DAGs (chains, diamonds, multi-parent merges, re-derivation cycles) shows 0 violations [internal, `a6_no_mint.py`, seed 20260816], while the copy-full mutant — forks inherit the undebited record — mints in a 2-operation shortest crime and an 8-way copy-fork multiplies one witnessed episode into 8.2× its quorum weight. One wrong turn on record: the budget-only phrasing (per-outcome grant budgets $\sum w \leq 1$, no debiting) was swept before proving and *refuted* — a full-weight chain at $\gamma = 0.9$ is budget-legal at every hop yet its closure sums to $0.9 + 0.81 + 0.729 = 2.439 > 1$ [verified: three terms of a geometric series, checkable by hand], so budgets without debiting conserve nothing for $\gamma > \frac{1}{2}$; the transfer restatement is the theorem. *Now you try:* check by hand that under transfer semantics the same chain's live total is $0.9 \cdot 1 = 0.9 \leq 1$, not 2.439 (Exercise 5.10).

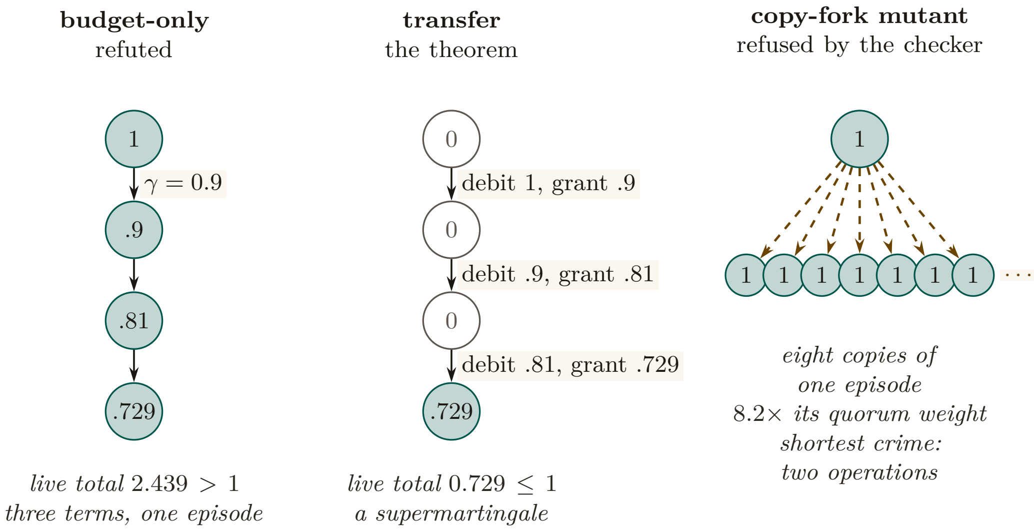


Figure 3: Three lineages of one witnessed episode of weight 1, derived at $\gamma = 0.9$. Under the budget-only phrasing every derivation stays live and the closure sums to $0.9 + 0.81 + 0.729 = 2.439 > 1$: reputation minted from nothing, which is why that phrasing was refuted before proving. Under transfer each derivation grants the child $\gamma\, w_p$ and debits the source w_p, so only the leaf is live and the total never exceeds what was witnessed. The copy-full mutant, which forks the undebited record, turns one episode into 8.2× its quorum weight in two operations, and the checker refuses it. [verified for the chain by hand; internal, `a6_no_mint.py`, seed 20260816]

Exercise 5.10, page 39.

Recall.

1. Without looking: what is the difference between connectedness and continuity, and which one does reputation require?

2. State in one sentence why "copy, not transfer" mints reputation.
3. What does the no-mint theorem conserve, and what is it silent about?

5 The three organs of continuity

"Continuity" is routinely used to mean three different things. Conflating them is the most common way a reputation design fails its own honesty test. To keep the argument concrete and honest, we measure each organ against a *reference system* — a running coordination daemon for local agent swarms whose component-by-component maturity is tabulated in Appendix A. Memory is implemented; checkpointing is partial; and the third organ has a shipped commitment-and-closure substrate but not yet the reputation-grade witnessed ledger this paper requires. We use the neutral maturity scale throughout (**implemented** / PARTIAL / SPECIFIED / *proposed*) and round nothing up.

Table 4: The three continuity organs have different survival depths. A successor inherits what each organ stores, and each stores less than the live body had: episodic memory keeps the notes whole, the checkpoint keeps a summary and the worktree lineage but not execution state, and the outcome ledger keeps commitments and closures only as far as they were witnessed. [internal]

Organ	What the successor inherits	What it does not
episodic memory	the notes, whole, with their ledger addresses	the execution state that produced them
checkpoint	a summary and the worktree lineage	the process's execution state; recovery restores notes, not execution
outcome ledger	the commitments c_1, c_2 and the witnessed outcomes o_1, o_2, o_3	an outcome nobody witnessed; a closure no oracle checked

5.1 Organ 1 — Memory: the episodic record implemented

An **episodic memory** component is the system's record of *what happened*: durable, typed episodes — each with a title, a summary, a source reference, and an agent/scope tag — promoted out of transient execution history. This design echoes the canonical agent-memory architecture of **Generative Agents** [5] (UIST 2023 — *a memory stream of natural-language observations, periodic* ***reflection*** *into higher-level inferences, and retrieval scored by recency* $\times$ *importance* $\times$ *relevance*): the promotion-of-episodes step is Park et al.'s importance gate by another name. In the reference system this organ runs in production; it is **implemented**.

5.2 Organ 2 — Checkpoint: restorable state partial

A checkpoint is *restorable execution/belief state*: enough that a successor *resumes* where the predecessor stopped. The reference system has a **restart organ** — a heartbeat-staleness detector that flags dead agents for salvage and republishes their unfinished work. It must be described formally, and it is: the restart organ **forwards a summary, not state**. It is checkpoint-*of-record*, not checkpoint-*of-execution*. The successor inherits a note about what the predecessor was doing, not the live belief state it was doing it with. The goal is sometimes called a "checkpoint with teeth" (the same phrase this paper uses everywhere else for it, §15, Appendix A); the current organ has gums.

Pitfall. Selling a summary-forwarding restart organ as real checkpointing would violate the system's own **honest-attestation** discipline: *only report "all good" when you have actually verified it — absence of error is not attestation.* Strong continuity — the kind that lets a successor *resume* rather than *inherit a summary* — has a specified candidate mechanism, **Event-Sourced Neural Rehydration** (OP-4, shared with *The Single-Writer Kernel*): restore the Git

SHA, truncate the durable message log to the exact crash point, and replay it under prompt-prefix caching. It is SPECIFIED, not **implemented**, and its *completeness has not been formally established.* What replay demonstrably restores is *lineage* — which commit, which claim, which message sequence the successor is continuing — and, *within one provider, model, and version*, a behaviorally equivalent continuation of the same prefix. What it does not restore is the model's actual inference state: a KV cache, sampling state, and hidden activations are bound to a live inference session and are not exported artifacts, so across providers (or across a model version bump) the right description is *task continuity with actor succession*, not resumption. The measurable form of that claim is a *fidelity rate*: the fraction of replayed runs that resume every pending obligation with no duplicated effect and no false belief, graded per provider and model version; until the mechanism ships it is an empirical hypothesis with a stated measurement, not a property. The formal-core result on the same seam says only that "the ledger followed the right body" — it makes no claim that the checkpoint restored the agent's experience, and neither do we. Until the mechanism ships and is measured, the restart organ still forwards a summary, and this chapter's own maturity ledger (Table 10) is where to check whether that has changed.

Co-ownership note (with the formal-core paper). The *kernel mechanism* of a checkpoint-with-teeth — the content-addressed execution snapshot — is owned by the formal-core chapter, *The Single-Writer Kernel*, which treats it as a low-level primitive. *This* paper owns the argument for *why it is the foundation of personhood and reputation.* The two papers state the same canonical sentence (next).

5.3 Organ 3 — Outcome ledger: the witnessed record of delivery partial

The third organ is the one reputation actually keys on. Its foundation is now PARTIAL: a durable commitment record, daemon-derived deadlines, oracle-bound closure, CLI/API surfaces, and an overdue-obligation monitor ship in the reference system. The complete organ remains an **append-only, daemon-witnessed outcome ledger** whose records are neutral, graded, identity-bound, and sanctionable. Its specified form is a **durable-commitment** record: a violable promise bound one-to-one to an actor, auto-enrolled when the actor takes a claim, closed only against an oracle, and watched by an obligation monitor that is the dual of the restart organ. A commitment row records what was promised; closing it requires an *oracle reference* — a released claim, a merged commit SHA, a passing test id, a satisfied policy sub-check. That commitment-and-closure seam ships. Neutral grading events, reputation updates, judge independence, sanctions, and durable identity binding do not; therefore the organ as a reputation ledger is partial, not complete.

Definition 5.1 (Oracle)**.** An **oracle** is a trusted source of ground truth that the agent being graded *cannot author*. An outcome is *witnessed* iff its closure references an oracle. The ledger is the Parfitian chain of §4 made concrete: a transitive sequence of witnessed deliveries that survives every respawn.

5.4 The canonical seam sentence (stated identically in every paper)

Key idea. The economy rests on three continuity organs — memory (implemented), checkpoint (partial: notes, not execution state), and the witnessed-outcome ledger (partial: commitments and oracle-bound closure, not neutral graded outcomes) — and reputation keys on the richer third organ, whose neutral, reputation-grade form is not yet complete; the checkpoint organ is the weakest continuity link and "checkpoint with teeth" (a real execution-state checkpoint) is the literal foundation of L3.

Table 10 is the honest-state ledger for this paper: every major claim with its label and its grounding. It is the artifact a skeptical reader should consult first.

Exercises 5.11–5.13, page 39.

6 Identity: the root the whole chain hangs from

Before any organ of continuity matters, the identity it attaches to must be one the agent cannot freely re-pick. This is the most under-appreciated claim in the paper.

In one breath. A reputation is worth exactly what it costs to walk away from; an identity that can be re-picked for free carries none, however long its record.

We first state the central claim of the section as a theorem and prove it, because it is the governing impossibility result the rest of the paper leans on: no amount of estimator cleverness can rescue a reputation built on a re-pickable identity.

Definition 6.1 (Sanction-respecting reputation)**.** [1] Fix a reputation mechanism M that maps each identity i to a score $r(i)$ used to gate economic access. Call M **sanction-respecting** if there exists a penalty $\Delta > 0$ such that whenever an actor produces a *dishonest* witnessed outcome (a delivery later overturned by an oracle, per Def. 5.1), the mechanism reduces the score available to that actor by at least Δ for some non-trivial window, relative to never having produced the outcome.

Theorem 6.1 Non-forgeable identity is necessary. If an actor can mint a fresh identity at cost $c = 0$ and freely route future work through it, then no mechanism M is sanction-respecting. Equivalently: a non-forgeable identity (one with strictly positive abandonment cost) is *necessary* for any sanction-respecting reputation.

Proof. Suppose M is sanction-respecting with penalty $\Delta > 0$, and suppose identity minting is free. Let an actor produce a dishonest outcome under identity i, so that by Def. 6.1 the score it can act under drops to at most $r(i) - \Delta$. Because minting is free, the actor instantiates a fresh identity i' at cost 0 and routes all future work through i'. By construction i' has no witnessed history, so M assigns it the newcomer score r_0, the same score any first-run identity receives. The actor's *effective* post-sanction score is therefore $\max(r(i) - \Delta,\ r_0) = r_0$ whenever $r(i) - \Delta < r_0$, and the actor incurs the sanction for at most the cost of switching identities, namely 0. The penalty Δ is thus not borne: the actor's accessible score after a dishonest outcome equals the score of a clean newcomer, with no window in which it is reduced below r_0. This contradicts Def. 6.1, which requires a reduction *relative to never having produced the outcome* — but a clean newcomer is, by definition, indistinguishable from "never having produced the outcome." Hence no sanction-respecting M can exist when $c = 0$. Contrapositively, $c > 0$ (a non-forgeable identity with positive abandonment cost) is necessary. □

Numbers by hand. The expression $\max(r(i) - \Delta,\ r_0)$ in the proof is one line of arithmetic; check it. Let the newcomer score be $r_0 = 50$ and let a dishonest outcome cost $\Delta = 30$.

[1] In the first edition this was Definition III.6.1; the standalone research paper on reputation inheritance cites it under that number.

Actor	$r(i)$	$r(i) - \Delta$	Accessible score	Does the sanction bite?
Mediocre record, free minting	50	20	$\max(20, 50) = 50$	No — it re-mints and is back at 50, exactly where the sanction was supposed to leave it worse off.
Strong record, free minting	90	60	$\max(60, 50) = 60$	Yes — abandoning costs it $60 - 50 = 10$, so it keeps the identity and eats the 30.

Table 5: Free identity minting does not defeat *every* sanction — it defeats every sanction that would push the actor below the newcomer floor r_0. That is the whole content of Theorem 6.1: the floor, not the penalty, is what an attacker optimizes against.

Row 2 is the more instructive one: an actor with a strong enough record has nothing to gain from whitewashing one bad outcome, because the escape hatch is capped at r_0. That does not weaken Theorem 6.1 — being sanction-respecting is a claim about *every* actor, and row 1 is a single actor for whom the penalty is not borne, which is all a counterexample needs. It does explain why the theorem is a *necessity* result rather than a security proof: what free minting destroys is the guarantee, not every instance of it. *Now you try:* at what prior score does the actor become exactly indifferent? ($r(i) = r_0 + \Delta = 80$; below that the mechanism is not sanction-respecting for that actor, above it the sanction is self-enforcing — and turning "how far above" into a designable quantity is exactly what Theorem 6.2 does next.)

Pitfall. Theorem 6.1 is a *necessity*, not a sufficiency, result: a non-forgeable identity is the gate, but it does not by itself make a reputation honest — you still need witnessed outcomes (§5) and an incentive-compatible grading oracle (§11). The theorem only rules out the cheap escape; it does not certify the score.

The dual fact — that free identities also defeat any *quorum* or *trust-aggregation* mechanism by replication — is the Sybil attack [7], and the same positive-cost requirement defeats it. We record both consequences as one property.

Property 6.1 (Reality of Reputation and Principal Liability)**.** A reputation system carries economic force only to the extent that the accountable principal cannot costlessly discard or multiply the liability-bearing identity. If an actor can freely obtain a fresh identity whose accessible economic utility (credit, exposure ceiling, opportunity) matches or exceeds its sanctioned utility, identity-local sanctions are evadable (**whitewashing**). If an actor can costlessly multiply identities to inflate votes or aggregate unreserved credit, quorum mechanisms collapse (**Sybil** [7]). Effective deterrence therefore requires binding all spawned actors to a durable principal with aggregated exposure limits.

Most agent runtimes still accept a *self-asserted string*: a human-readable triple of the form *project : stack : context* that the operator chooses and can re-pick at will. The reference system now ships the first local correction: `actor-souls` mints an opaque actor id and lookup credential inside the daemon, and the budget guard puts fresh actors into a shared, bounded newcomer pool. Legacy and bypass paths still accept asserted identifiers, and the credential is not yet enforced at every security-relevant write boundary. The result is a real but partial local identity root, with the asserted string demoted toward a display alias; it is not yet the complete identity contract this paper requires.

6.1 S2 — whitewashing in action

Return to Alice's swarm. Tonight's *Drake* deleted the failing test instead of fixing it; the daemon's outcome log records the deletion and a later re-audit flags it. On a legacy or bypass path that still accepts self-asserted identity, the sanction is free to escape: when Alice spawns tomorrow's swarm, she (or Drake's own boot prompt) registers the agent as *project : stack : context2*, and a brand-new identity inherits a *clean slate*. The bad record attaches to a string nobody is forced to keep. This is whitewashing, and it is not exotic: it is the default behavior of any re-pickable identity. Now replay the same night with a non-forgeable identity. The re-audit's tombstone (§12) attaches to the credential Drake *cannot* drop; tomorrow's incarnation either presents that credential — and carries the sanction — or presents a fresh one and is treated as a *newcomer* with a reduced economic ceiling (below). Either way the sanction is no longer free, which is the whole point: the identity must cost more to abandon than a clean record is worth. We make that "costs more" precise in Theorem 6.2.

6.2 The classic attacks this forecloses

- **Cheap-pseudonym whitewashing** [6]: if a fresh identity is free, a bad record is shed by respawn. The defense is a *newcomer policy as a shape, not a scalar*: a newcomer may have full *work* ability but a reduced *economic ceiling* until it has a witnessed history — so identity churn costs more than a clean record is worth.
- **The Sybil attack** [7]: free identities defeat any reputation or quorum mechanism by replication. The defense is a daemon-minted id whose creation is rate-limited and, in the economy, *bonded*.

Theorem 6.1 says abandonment must cost *something*; the design question is *how much*. The newcomer policy answers it as a *shape, not a scalar*, and that shape implies a clean bound on what a whitewasher pays.

Theorem 6.2 Whitewashing-cost bound. Let an honest actor's witnessed history raise its economic ceiling from the newcomer ceiling κ_0 to a steady-state ceiling $\kappa^\star > \kappa_0$ over a maturation window, and let the per-period surplus an actor can extract be non-decreasing in its ceiling, with surplus $\pi(\kappa)$. Suppose an identity, once sanctioned, can only re-enter as a newcomer at ceiling κ_0. Then an actor who whitewashes (abandons a sanctioned identity and re-enters fresh) forgoes at least

$$W \;=\; \sum_{t=1}^{T} \big(\pi(\kappa^\star) - \pi(\kappa_t)\big) \;\geq\; 0,$$

the cumulative surplus gap over the maturation window T during which the fresh identity climbs from κ_0 back toward $\kappa^\star$. Whitewashing is unprofitable exactly when the one-time gain from escaping the sanction is less than W; choosing the ceiling schedule so that W exceeds the maximum single-outcome fraud gain makes honesty the dominant strategy for any actor that intends to keep working.

Proof sketch. A sanctioned identity that does not whitewash retains its ceiling $\kappa^\star$ (minus the explicit sanction); an identity that whitewashes resets to κ_0 and, by hypothesis, can only regain the ceiling by re-accumulating witnessed history over T periods, earning $\pi(\kappa_t)$ in period t with $\kappa_0 \leq \kappa_t \leq \kappa^\star$. The opportunity cost of the reset is the per-period surplus gap summed over the window, $W = \sum_t (\pi(\kappa^\star) - \pi(\kappa_t))$, which is non-negative since π is non-decreasing in the ceiling. An actor whitewashes only if the avoided sanction exceeds W; setting the schedule $\{\kappa_t\}$ (equivalently, the maturation rate) so that W dominates the largest gain a single fraudulent outcome can capture removes the incentive. The key design move is that the newcomer gets *full work ability* but a *reduced economic ceiling*, so W is paid in forgone *earnings*, not in forgone *ability* — which is why it taxes whitewashers without taxing honest newcomers' capacity to contribute. □

Numbers by hand. W is a sum you can do on paper. Set the newcomer ceiling $\kappa_0 = 10$, the steady-state ceiling $\kappa^\star = 100$, a linear five-period climb $\kappa_t = 10 + 18t$, and per-period surplus $\pi(\kappa) = 0.1\kappa$.

Period t	1	2	3	4	5	**total**
Ceiling κ_t	28	46	64	82	100	—
Surplus $\pi(\kappa_t)$	2.8	4.6	6.4	8.2	10.0	—
Gap $\pi(\kappa^\star) - \pi(\kappa_t)$	7.2	5.4	3.6	1.8	0.0	$\mathbf{W = 18.0}$

Table 6: The whitewasher's bill, computed. Re-entering as a newcomer forgoes $W = 18.0$ of surplus over the maturation window; any single-outcome fraud gain below that is not worth the reset.

So a fraud gain of $G_{\max} = 15$ is deterred by this schedule ($15 < 18$) and a gain of $G_{\max} = 25$ is not ($25 > 18$) — the design lever is the *shape* of the climb, not the size of the penalty. Note also what this instance fixes for the next theorem: the largest holdback any single period can carry here is $L = \pi(\kappa^\star) - \pi(\kappa_0) = 10 - 1 = 9$, and that L is precisely the per-period ceiling that Theorem 6.3 must respect. *Now you try:* stretch the same endpoints over ten periods instead of five — what is W? ($\kappa_t = 10 + 9t$, so the gaps are $9 - 0.9t$ and $W = 90 - 0.9 \cdot 55 = 40.5$: doubling the window more than doubles the bill, because the early gaps are the expensive ones.)

Key idea. The newcomer policy is a *shape*: full work ability, reduced economic ceiling, climbing with witnessed history. That shape is what makes Theorem 6.2's cost W fall on whitewashers (who keep resetting their ceiling) and not on honest newcomers (who keep theirs and climb). A flat "distrust all newcomers" policy taxes the wrong party.

Theorem 6.3 Front-loaded probation dominance, under a ceiling. Let a newcomer restriction schedule specify earning holdbacks g_t across periods $t \in \{0, 1, \ldots, T\}$. Let honest participants have patient discount factor $\delta_h \in (0, 1)$ while strategic whitewashers (who seek short-term defection gains $G_{\max}$) have impatient discount factor $\delta_f < \delta_h$. Because g_t is the amount by which a newcomer's economic *ceiling* is reduced, it is bounded above by that ceiling: the schedule is constrained by

$$0 \;\le\; g_t \;\le\; L, \qquad L \;:=\; \pi(\kappa^\star) - \pi(\kappa_0),$$

the per-period cap supplied by Theorem 6.2's own ceiling schedule. Subject to the deterrence constraint $\sum_{t=0}^{T} \delta_f^t g_t \;\ge\; G_{\max}$, the schedule minimizing honest newcomers' lifetime friction $H(g) = \sum_{t=0}^{T} \delta_h^t g_t$ is **bang-bang, filled from** $t = 0$: there is an index k with $g_t = L$ for $t < k$, $g_k \in (0, L]$, and $g_t = 0$ for $t > k$ — a cliff of positive *width*, collapsing to the pure spike $g^\star = (G_{\max}, 0, \ldots, 0)$ with $H(g^\star) = G_{\max}$ exactly when the cap does not bind ($G_{\max} \le L$). Any gradual ramp that defers mass to later periods is strictly dominated. Two qualifications bind and are part of the statement, not footnotes to it:

1. **The pure spike is infeasible whenever $G_{\max} > L$.** Front-loading survives; *uniqueness* and the closed form $H(g^\star) = G_{\max}$ do not.
2. **The whole programme is infeasible when $G_{\max} > L/(1-\delta_f)$.** Total achievable deterrence is bounded by $\sum_{t\ge 0} \delta_f^t L = L/(1 - \delta_f)$, so past that point *no* schedule shape deters, and the honest response is a different instrument (a bond, an escrow), not a different schedule.

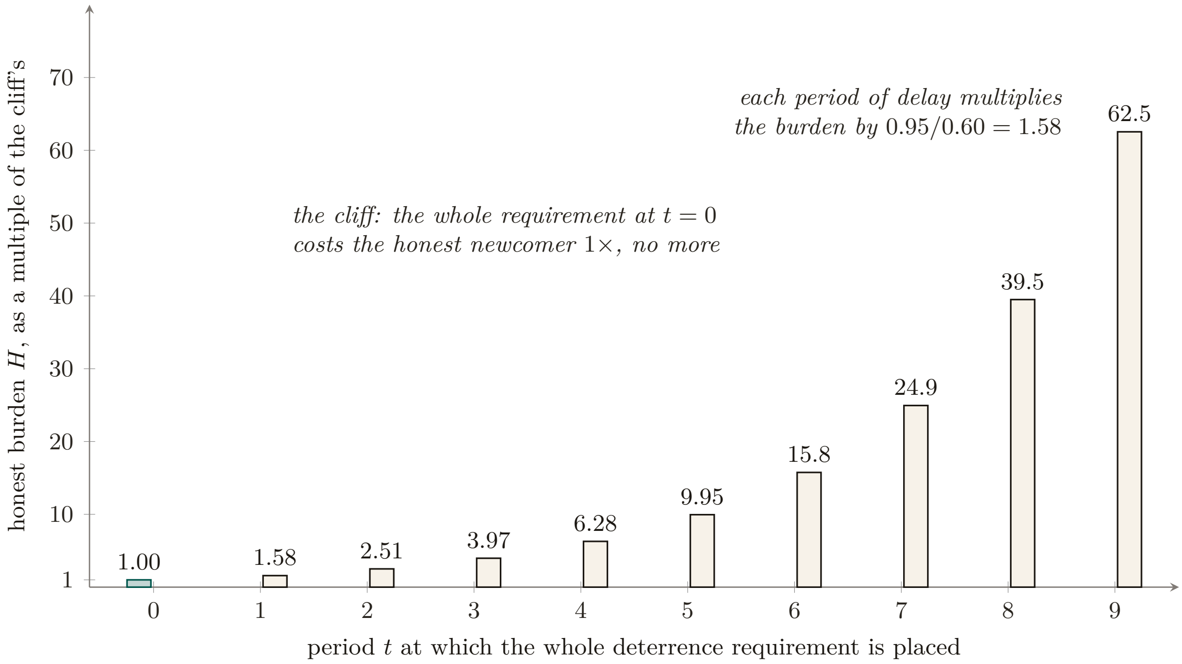


Figure 4: The honest newcomer's burden of one unit of deterrence, by the period in which it is placed, at $\delta_h = 0.95$, $\delta_f = 0.60$. Deterrence against a whitewasher discounts at δ_f while the honest newcomer discounts at δ_h, so a restriction placed at period t costs the honest party $(\delta_h/\delta_f)^t$ times what the same deterrence costs at $t = 0$: $1.58\times$ after one period, $62.5\times$ after nine. The front-loaded schedule, the cliff, is undominated; the random search over 4,000 instances found no schedule below it. [internal; `b6_probation.py`, seed 20260816]

Proof sketch. Minimize $H(g) = \sum_t \delta_h^t g_t$ subject to $\sum_t \delta_f^t g_t \geq G_{\max}$ and $0 \leq g_t \leq L$.

The exchange step, done correctly. The tempting move — "shift mass from a later period to period 0" — is wrong as usually stated, and the error is worth naming because it inverts the sign. Shifting mass m one-for-one from period t to period 0 changes the honest cost from $\delta_h^t m$ to m, and since $\delta_h < 1$ that is an *increase*, not a decrease; what the move buys is slack in the deterrence constraint, which is not the objective. The valid exchange holds deterrence *tight*. Take a feasible g with $g_t = m > 0$ for some $t \geq 1$ and $g_0 < L$. Reduce g_t by $\varepsilon \in (0, m]$ and raise g_0 by $\delta_f^t \varepsilon$ (choosing ε small enough that $g_0 + \delta_f^t \varepsilon \leq L$); call the result g'. The deterrence sum is unchanged, since it loses $\delta_f^t \varepsilon$ at period t and gains $\delta_f^0 \cdot \delta_f^t \varepsilon = \delta_f^t \varepsilon$ at period 0. The honest cost, meanwhile, changes by

$$H(g') - H(g) \;=\; \delta_f^t \varepsilon \;-\; \delta_h^t \varepsilon \;=\; -\,\varepsilon\big(\delta_h^t - \delta_f^t\big) \;<\; 0 \qquad \text{for every } t \geq 1,$$

strictly, because $\delta_h > \delta_f$ makes $\delta_h^t > \delta_f^t$. So deterrence-preserving mass movement toward $t = 0$ strictly lowers the honest tax. Equivalently: a unit of deterrence bought at period t costs the honest newcomer $\delta_h^t / \delta_f^t = (\delta_h/\delta_f)^t$, strictly increasing in t, so deterrence should always be bought at the cheapest available date.

From the exchange step to the shape. Iterating fills period 0 to the cap L before any mass sits at $t = 1$, then period 1 before any sits at $t = 2$, and so on; the process halts at the first index k where the accumulated deterrence $\sum_{t<k} \delta_f^t L + \delta_f^k g_k$ reaches $G_{\max}$. That is exactly the bang-bang fill-from-zero shape claimed, and it is an optimal vertex of the LP for *every* $\delta_f < \delta_h$ — the corner is not a knife-edge.

The uncapped closed form. If $L \geq G_{\max}$ the cap never binds, $k = 0$, and for every feasible g,

$$H(g) \;=\; \sum_t (\delta_h/\delta_f)^t \, \delta_f^t g_t \;\geq\; \sum_t \delta_f^t g_t \;\geq\; G_{\max} \;=\; H(g^\star),$$

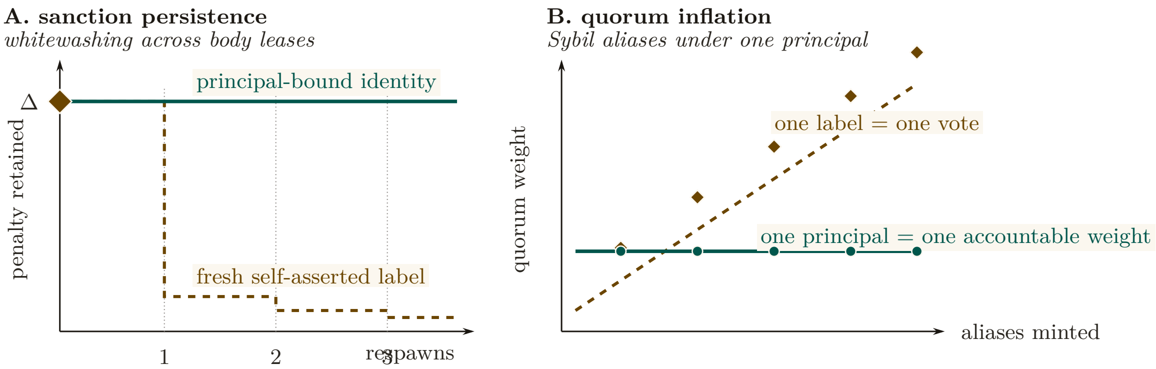
with equality only when all mass sits at $t = 0$: the spike is then the unique minimizer.

Infeasibility. With $g_t \le L$ the largest deterrence any schedule can deliver is $\sum_{t=0}^{T} \delta_f^t L < L/(1-\delta_f)$, so the constraint set is empty once $G_{\max} > L/(1-\delta_f)$ — at *every* schedule shape. Hence, among all schedules the ceiling can actually carry, the one that minimizes honest newcomers' lifetime friction is a front-loaded probation cliff followed by graduation to full capacity. □ □

Numbers by hand. Take $\delta_h = 0.95$, $\delta_f = 0.60$, $G_{\max} = 20$. *Cap slack ($L \ge 20$):* the cliff is the pure spike and costs the honest newcomer exactly 20; holding the same deterrence with all the mass at $t = 5$ instead costs $20 \cdot (0.95/0.60)^5 = 199.0$ — ten times the honest tax for zero extra deterrence. *Cap binding ($L = 12$):* feasible, since $L/(1-\delta_f) = 12/0.4 = 30 \ge 20$. Fill from zero: $g_0 = 12$ (deterrence 12), $g_1 = 12$ (deterrence $0.6 \cdot 12 = 7.2$, running total 19.2), and the residual 0.8 at $t = 2$ needs $g_2 = 0.8/0.36 = 2.22$. Honest cost $H = 12 + 0.95 \cdot 12 + 0.9025 \cdot 2.22 = 25.41$ — the cliff has acquired *width*, and the ceiling has cost honest newcomers 5.41 over the uncapped optimum. *Now you try:* at $L = 8$, is the schedule feasible at all? ($L/(1-\delta_f) = 8/0.4 = 20 = G_{\max}$ — feasible only in the degenerate limit of an infinite horizon held at the cap forever; any $L < 8$ makes this fraud gain undeterrable by probation alone, and you need a bond instead.)

Numbers by hand — The probation cliff: falsifying the schedule shape. This is the whitepaper statement of the companion paper's probation-cliff theorem [31], carrying its amended ceiling hypothesis. Exchange argument verified numerically across randomized ($\delta_f < \delta_h$, $G_{\max}$) instances: 0 dominating schedules in 4,000 draws — the number regenerates from `b6_probation.py` at seed 20260816, testing 76,000 candidate schedules with the deterrence constraint held tight, and the exchange step's cost ratio $(\delta_h/\delta_f)^t$ verified strictly monotone in t [internal, `b6_probation.py`, seed 20260816]; the polished LP write-up is carried as item B6 of the research ledger. The winning shape is a **cliff**: harshness held at the ceiling L from $t = 0$, dropping to 0 the moment accumulated deterrence clears $G_{\max}$, never eased in gradually — any schedule that defers mass to a later period is strictly dominated, so the corner solution is the whole answer, not an approximation to one. **Boundary on the evidence:** that sweep was generated *without* the per-period cap, so it certifies the uncapped statement and the exchange step, not the bang-bang shape under a binding L; re-running it with $g_t \le L$ is an open verification debt. *One citation is owed and unresolved:* the foil above is stated as the deferred-compensation folklore one might import from Lazear, not as a confirmed reversal of his result. If his contract front-loads the honest worker's implicit bond in the same direction, the cliff agrees with him rather than correcting him. Here the re-payable hurdle is itself the punishment a whitewasher faces on every reset, so front-loading taxes resets, not tenure — but that framing stays folklore until the citation is checked. *Now you try:* run the script yourself and check the instance count against the 76,000 figure above (Exercise 5.17).

Figure 5 draws the two attacks a free identity enables — whitewashing a sanction and Sybil-multiplying a quorum — and the non-forgeable root that closes both.



scope: one trusted daemon and one bound principal; federation still owes cross-operator attestation

Figure 5: Non-forgeable identity changes the economics of both classic attacks. In panel A, a self-asserted label drops a sanction by respawning; a principal-bound identity carries the penalty Δ across every body lease. In panel B, free labels turn aliases into quorum weight, while principal binding keeps one accountable weight constant. The claim is deliberately local: daemon minting and principal binding close these attacks inside one trusted domain; they do not solve cross-operator attestation. Newcomer friction applies to economic ceiling, not to the ability to perform work.

6.3 The principal above the actor

The economic counterparty in every trade is not the agent but the **principal** — the human or organization that owns the fleet and stands behind its bonds. Bonds, reputation inheritance, and sanctions must ultimately key on the principal via the delegation chain, or Sybil bond-farming works by spawning fresh *agents* under one principal. This paper defines the principal as the delegation chain's terminus and the identity object on which the market paper's counterparty is built; the market paper *cross-references* this definition rather than redefining it.
The structure is old enough to have a canonical statement. Hobbes's chapter on persons [28, ch. XVI] separates the *actor* — whose words and actions are performed — from the *author*, whose they are *owned* by, and locates liability at the author: an artificial person acts, and someone else is answerable for it. That is precisely the split this paper needs between a spawned agent and its principal, and it carries the same warning: an actor with no identifiable author is an actor no counterparty can hold to anything.

Definition 6.2 (Principal)**.** A **principal** is the root of a delegation chain: the durable economic entity (human/org) that owns a fleet, posts its bonds, and inherits the reputational consequence of its agents' witnessed outcomes. Each agent's non-forgeable identity is bound to exactly one principal at mint time.

6.4 The body behind the name: engine swaps and resurrection

Everything above binds a record to a name the actor cannot re-pick. It says nothing yet about what is running *behind* the name.[2] Two operations that are routine for software and exotic for people force the question. An identity can swap the model engine behind its name between one witnessed outcome and the next, and an identity whose provider dies can be resurrected from a checkpoint somewhere else. Both are body changes under a constant name, and the ledger must decide what follows the name and what follows the body.

In one breath. Unattested, a price can never depend on the engine actually running, so swapping a cheap engine in behind a good record pays at every price and pooling on quality dies by Akerlof's spiral inside a single identity. Attest the engine on every witnessed outcome and,

[2] A standalone, submission-form version of this section is *Continuity Without Metaphysics* (research paper 5) [31]; the sweeps and the migration model check cited here are its artifacts, regenerated by `b5_engine_substitution.py` at seed 20260816.

provided the attested price passes the full quality difference through, the substitution incentive flips to the planner's own efficiency rule at zero audit stake. A record may then cross providers exactly when three checks pass together, and dropping any one of them opens a named two-step attack.

The scene. An identity builds a glowing record running a strong engine θ_H, then quietly runs a cheap engine θ_L behind the same name. Buyers see the reputation; they do not see the engine. This is the used-car market of Akerlof [24] with one twist that makes it sharper: the asymmetric information is not about which seller you drew but about what is running behind a name you already trust. The lemon is wearing the plum's history. Reputation was supposed to solve adverse selection by aggregating past quality [25]; the swap severs the link between the record and the machine that will produce the next outcome, so the record measures the wrong object.

Theorem 6.4 Adverse selection within one identity, and the attested flip. Let two engines $\theta_H > \theta_L$ run at costs $c_H > c_L$, write $\Delta\theta = \theta_H - \theta_L$ and $\Delta c = c_H - c_L$, and let a buyer pay p for a witnessed outcome.

(a) *Unattested.* If the engine is unobserved, the one-period gain from swapping, $(p - c_L) - (p - c_H) = \Delta c > 0$, is independent of p: no price supports pooling on θ_H, and no costless signal inside the identity separates the types. Committed θ_H sellers stay in the market exactly when the pooled price clears their cost, $\mu\theta_H + (1 - \mu)\theta_L \geq c_H$ (the model's participation condition, not a theorem), that is when the committed share satisfies $\mu \geq \mu^\star = (c_H - \theta_L)/\Delta\theta$; such a share exists only if $c_H \leq \theta_H$, since $c_H > \theta_H$ gives $\mu^\star > 1$ and no committed share supports the strong engine at any price. Below $\mu^\star$ the spiral runs inside one identity down to θ_L. Deterring the swap without attestation needs an audited bond with $qB \geq \Delta c$ (audit probability q, slashable bond B).

(b) *Attested.* If the daemon attests the engine id on every witnessed outcome, reputation keys on the pair (principal, engine), the price schedule splits into p_H and p_L, and the swap gain becomes $\Delta c - (p_H - p_L)$. If the attested schedule prices quality at its full social value, $p_H - p_L = \Delta\theta$, the gain is $\Delta c - \Delta\theta$: swapping pays exactly when the cheap engine is socially efficient, which is the planner's own rule, at zero audit stake.

The pass-through condition in (b) is a hypothesis about the market, not a consequence of attestation: under Bertrand competition on attested keys $p_\theta = c_\theta$ and the gain collapses to zero, and any schedule that passes through less than the full quality difference moves the flip point.

Proof. (a) The buyer's payment does not depend on the engine, so the seller's payoff difference between the two engines is the cost difference alone. A gain that is constant in p cannot be priced away, and a signal that costs the low type nothing more than the high type separates nobody. The pooled price at committed share μ is the expected quality $\mu\theta_H + (1 - \mu)\theta_L$; a committed seller whose cost exceeds it exits, which lowers μ and the price after it, and that descent is the spiral. An audit that catches the swap with probability q and slashes B makes the expected gain $\Delta c - qB$, nonpositive iff $qB \geq \Delta c$. (b) With the engine attested, a seller receives the price of the engine it ran, so the payoff from swapping is $(p_L - c_L) - (p_H - c_H) = \Delta c - (p_H - p_L)$. Under full pass-through this is $\Delta c - \Delta\theta$, and $\Delta c > \Delta\theta$ says precisely that $\theta_L - c_L > \theta_H - c_H$: the cheap engine produces more surplus. No audit enters, because the gain is already nonpositive whenever the strong engine is the efficient one. □

Checked. Sweeps at seed 20260816: 4,000 instances $\times$ 8 prices with 0 violations of price-independence; the $\mu^\star$ and qB lines match simulation with 0 mismatches; 2,000 heterogeneous-cost spirals are all monotone to the Akerlof fixed point ($\mu_0 = 0.5 \to \mu_\infty = 1/3$ at price 0.6; $\mu_0 = 0.2 \to 0$ at price $\theta_L = 0.4$); the attested incentive sign equals the efficiency sign in 4,000/4,000 draws [internal, `b5_engine_substitution.py`]. The wrong-threshold mutant $\mu^\star = \Delta c/\Delta\theta$ is caught 1,305 times and the no-flip mutant (attestation without re-keying the price) 1,975 times [internal].

Numbers by hand — The swap gain and the attested flip. Take $\theta_H = 1$, $\theta_L = 0.4$, $c_H = 0.5$, $c_L = 0.2$. Unattested, the swap gain is $c_H - c_L = 0.3$ at every price [verified]. The pool clears c_H when $\mu \cdot 1 + (1-\mu)\cdot 0.4 \geq 0.5$, that is $0.6\mu \geq 0.1$, so $\mu^\star = 1/6 \approx 0.167$ [verified]; the deterrence bond at audit rate $q = 0.6$ is $B^\star = \Delta c/q = 0.5$ [verified]. Attested, keeping the strong engine earns $\theta_H - c_H = 0.50$ and swapping earns $\theta_L - c_L = 0.20$, so the flip is $\Delta c - \Delta\theta = 0.3 - 0.6 = -0.3$ and the required audit stake is 0 [verified; the sweep reproduces every line, internal]. *Now you try:* raise the strong engine's cost to $c_H = 0.7$. What committed share keeps the unattested pool alive? ($\mu^\star = (0.7-0.4)/0.6 = 0.5$: half the market must be committed before quality can break even [verified].) Then run the script's Part 3 (Exercise 5.18).

What it buys. The buyer-facing sentence in its provable form: reputation you can trust because the engine behind it is attested. The misread to preempt is that attestation forces everyone onto the expensive engine. It does not. Swapping pays exactly when $\Delta c > \Delta\theta$, when the cheap engine really is the better social bargain, so attestation does not freeze technology choice; it aligns it. What dies is only the trade that was profitable because it was invisible. The engineering literature on covert model substitution stops exactly where part (b) starts: Cai et al. [26] formalize substitution in commercial model APIs, show that software-only auditing is query-intensive and defeated by inference nondeterminism, conclude that hardware-backed attestation is the only robust channel, and name weak provider incentives as the obstacle to adopting it. The flip removes that obstacle. Attestation makes substitution unprofitable precisely when it is inefficient, so a provider's reason to attest is price rather than civic-mindedness.

Resurrection with teeth. The provider hosting an agent shuts down. The principal holds a checkpoint and a competitor will run it. Whether the resurrected process *is* the same agent is the question this chapter refuses; the question it answers is which verifications make it safe for the successor to inherit the predecessor's score, in the precise sense that Definition 6.1 still holds after the move. Resurrection must carry the debts, not just the assets.

Model-checked property 6.5 Resurrection soundness. Sanction-respecting reputation (Definition 6.1) is preserved across provider migration under three clauses: (i) the successor's lineage is verified by credential and continuity witness; (ii) every successor outcome carries an engine attestation, so the migrated score stays keyed to the attested engine that earned it; (iii) open commitments are closed or renegotiated with an escrow delta before cutover. Each dropped clause admits a shortest-trace escape. Without (i) a sanctioned identity re-enters clean through migration (*whitewash-by-resurrection*, two steps). Without (ii) the successor runs θ_L while being paid on the θ_H key, its accessible score exceeding even a clean twin's (*engine-history shedding*, two steps). Without (iii) in-flight obligations land on nobody (*liability stranding*, two steps).

Scope lemma. Prices key on the pair (principal, engine); sanctions key on the principal alone, across all of its engine keys. Collapsing either direction reopens a hole: engine-keyed sanctions hand over the sibling engine key inside the same identity as an escape hatch (caught in one step, no migration needed), and principal-keyed prices reopen part (a) of Theorem 6.4.

Checked. Bounded model check of the migration protocol: 747 reachable states to depth 7 (467 of them post-migration), and the intact protocol satisfies Definition 6.1 in every state. Mutants caught with shortest crimes: key-scoped sanctions in 1 step, whitewash-by-resurrection in 2, engine-history shedding in 2, liability stranding in 2 [internal, `b5_engine_substitution.py`].

The wrong turn. A first pass keyed sanctions the way it keyed prices, per (principal, engine), and the intact protocol failed Definition 6.1 in one step with no migration involved: work dishonestly on one engine key, keep acting at full score on the sibling key. The fix is the scope lemma's one line, price per (principal, engine) and sanction per principal, and it is the same rule that closes skill-versioning's escape (a sanctioned `v1` must not act clean through `v2`): one lemma, two doors. The resume button and provider migration therefore ship with their own security property and the three attacks it was tested against, by name. The composition with the no-mint law of §4 is exact: migration moves a score whole, with its sources closed, and forking splits one with a debit. Both are the same conservation law, so an adversary cannot arbitrage between the resurrection door and the fork door.

> **Where this stops.** The engine-swap theorem is a two-period stylization: the strategic type has no reputational continuation value beyond the bond B, and a long-horizon analysis would shift the unattested deterrence line. Attestation soundness is a deployment assumption (enclave and daemon integrity); the theorem prices what attestation buys, not whether a given attestation is sound. Quality θ is a scalar; multi-dimensional quality with task-dependent rankings needs the pricing key extended, not the sanction key. The migration result is bounded small-model checking of the protocol specification (depth 7, at most two migrations, integer scores); it certifies that the clauses have teeth, not that the clause list is complete. Definition 6.1 is score-only, so the escrow side of clause (iii) enters as attribution (the default lands on an identity), not as a monetary conservation theorem. Continuity witnesses attest lineage, not welfare: nothing here says the checkpoint restored the agent's experience, only that the ledger followed the right body.

Exercises 5.14–5.18, page 39.

Recall.

1. Without looking: what does a non-forgeable identity make necessary, and what does it *not*, by itself, certify?
2. Is the whitewashing-cost bound's design lever the size of the penalty or the shape of the ceiling schedule?
3. Why must a resurrection/migration protocol sanction by principal rather than by (identity, engine)?
4. Under attestation, when does an engine swap still pay, and why is that the intended outcome rather than a failure of the mechanism?

7 The keystone split: local identity vs. cross-operator attestation

The non-forgeable identity of §6 is now the highest-leverage *partial local* keystone of the program. The cross-operator stone remains unbuilt. The keystone is *two stones*, and conflating them is the single most consequential error the series can make.

Definition 7.1 (Local non-forgeable identity)**.** A **local non-forgeable identity** provides an unforgeable actor identity *within one trusted daemon*: the runtime issues and binds it, and no actor inside the fleet can re-pick or impersonate it. Its threat model is explicitly intra-fleet and single-operator; it is *not* a public-key infrastructure and offers no defense against a hostile *operator*. The reference implementation's daemon-minted actor-soul and credential satisfy a bounded slice of this definition; full write-boundary enforcement and migration of legacy identity paths remain. This is exactly what the single-operator layers—every chapter up to and including this one—need and assume. PARTIAL.

Definition 7.2 (Cross-operator attestation)**. Cross-operator attestation** is the *unsolved* problem of binding identity keys across harbors you do *not* own: an actual key-distribution and attestation story between mutually-distrusting operators. A federated witness log gives log-*integrity*, not identity-*binding* — it can prove an entry was recorded, but not that the key behind it is who it claims to be across a trust boundary. This is what the market paper needs and does *not* yet have. *proposed* (a named gap, owed to the market paper).

> *This paper depends on a* local non-forgeable identity *and* hands the market paper *the unsolved* cross-operator attestation *as a named dependency — not an assumption. Every cross-operator sentence that says "the boundary neither operator controls" is resting on cross-operator attestation; until that exists, it is resting on a single trusted root, and we say so.*

Table 7: The keystone split. Everything Alice's daemon knows about itself is also checkable from Bob's foreign view except the last row: a signed, intact history still does not prove who controls the foreign key. Log integrity crosses the trust boundary; accountable-principal binding does not, and that missing row is the stone the market rests on and does not yet have. [internal]

Claim	Inside Alice's trusted daemon	From Bob's foreign view
the actor key was minted by the daemon	known: the daemon minted it	the key remains visible; the signature checks
the history is append-only	known: the daemon wrote it	log integrity remains verifiable; inclusion proofs check
the registry is local	known	the foreign registry is inspectable, not trusted
the key is bound to an accountable principal	known: the daemon's own binding	**no cross-operator binding proof**: nothing Bob can check says who controls the key

Key idea. The keystone that holds up *this* paper (identity inside one trusted daemon) is *not* the keystone that holds up the cross-operator market. This paper stands on a local non-forgeable identity. The market paper needs cross-operator attestation and must declare it *proposed* rather than borrow the single-operator threat model. This is the clean boundary the series' cross-layer review demanded.

The second stone's envelope has been standardized — the stone has not. While this series was being written, the agentic-payments industry shipped the credential dialect the second stone will be carved in: the Agent Payments Protocol [33], adopted by the Universal Commerce Protocol [32] (Google/Shopify, 2026), expresses principal authorization as W3C Verifiable Credentials in SD-JWT form.[3] The reference implementation adopts this envelope at the harbor boundary (ADR-0094), and the honesty rider matters here more than anywhere: *a standard format shrinks the problem; it does not solve it.* What the profile buys is that a principal mandate verifies with off-the-shelf tooling, that principal keys can live in secure enclaves, and that key *distribution* reduces to fetching a cacheable HTTPS document. What it does not buy is the binding of Definition 7.2: a JWK set authenticates keys to an *origin*, not to a principal, and origin-binding across mutually-distrusting operators is exactly the unsolved attestation problem

[3] SD-JWT+kb with hardware-bindable ES256 keys for the principal, JWS detached-content signatures over JCS-canonicalized payloads for the counterparty, and keys published as JWK sets under a `/.well-known` profile.

— exercise (3)'s "shared root of trust" in its 2026 industrial form. Cross-operator attestation remains *proposed*; it is now *proposed* with a standardized serialization, which is progress of the unglamorous kind.

Exercises 5.19–5.21, page 39.

8 Reputation as a substrate, not a score

Here is the hinge of the whole series, stated as the canonical cross-paper sentence:

> *The reputation* estimator is cheap*; the* substrate it scores over *— witnessed outcomes on a non-forgeable identity — is the gate.*

A team tempted to start an agent economy by "building an Elo leaderboard for backends" would be building the cheap, last part first. Elo, Bradley–Terry, and TrueSkill are well-understood; you can implement any of them in an afternoon over a table of game results. The hard, prior work is the three links *beneath* the estimator: a non-forgeable identity (§6), continuity (§5), and an *oracle-witnessed* outcome ledger (Def. 5.1). Without those, the estimator scores noise — or worse, scores a number an adversary controls.

8.1 The estimator family (the well-understood part)

We catalog the estimators not because they are hard but because choosing the *wrong* one for the signal type is a real error.

- **Pairwise / tournament signal → Bradley–Terry / Elo / TrueSkill.** When the witnessed signal is "A beat B on this task," the **Bradley–Terry** model [8] is the full-history MLE, and **Elo** [9] is its online special case. A harbor that retains the *whole* history of every settled outcome lives in the Bradley–Terry regime, not the streaming-Elo regime; BT gives more stable ratings than Elo's recency-weighted update when the population is static. **TrueSkill** [10] adds a calibrated uncertainty σ — the principled way to say "we have not tested this backend much yet," which matters at cold start.
- **Trust-propagation signal → EigenTrust.** When the signal is "who does the network trust," **EigenTrust** [11] computes a global trust vector as the principal eigenvector of the local-trust matrix — the reputation-systems analogue of PageRank, and the canonical answer to "aggregate peer opinions while resisting collusive cliques." It is the right tool for the *federated* reputation *The Harbor Economy* needs (trust that travels across harbors), and it is *not* a bandit.
- **Marketplace feedback signal → feedback systems.** Resnick–Zeckhauser [12] measured a ∼8% reputation price premium on eBay; Tadelis [13] frames feedback systems as the platform's answer to adverse selection. This is the empirical evidence that the third side of the market has real value to price — and the reason a harbor is a *platform* problem rather than a scoring problem: Rochet–Tirole's two-sided-market analysis [23] shows that the price and quality signals a platform exposes to one side determine participation on the other, which is exactly the coupling a reputation substrate creates between the agents being rated and the buyers reading the rating.

8.2 Score as telemetry; gate on predicates

Key idea. Publish the score as *telemetry* (a signal the operator and the buyer can read), but *gate* on *predicates* (machine-checkable witnessed facts: "passed the acceptance test," "no slashed bond in the last k settlements"). The instant the score itself becomes the gate, it becomes a Goodhart target [19]: "when a measure becomes a target it ceases to be a good measure."

Exercises 5.22–5.24, page 40.

9 Why reputation is *not* a bandit problem

It is tempting — and the routing literature invites the temptation — to model "which agent, model, or skill should I pick?" as a **multi-armed bandit** [15]: arms are agents, pulling an arm yields a reward, and Thompson sampling [16] or UCB balances exploration against exploitation. That framing is correct for one narrow internal task and catastrophically wrong for reputation as a whole. The distinction is important enough to state as a claim.

Honesty rider before the claim. The four-assumption argument below is a **design argument, not a theorem**: unlike §6's necessity and whitewash-cost results or §11's imported contraction theorem, it has no numbered formal counterpart in any companion paper of this series. It is stated in the confident register because we believe it, and because getting it wrong is expensive — not because it has been proved.

> Internal routing — *one operator privately choosing which of their own agents to spawn next — is bandit-shaped.* Reputation — *a public, adversarial, multi-dimensional, persistent substrate other parties trade on — is not. Modeling reputation as a bandit imports four assumptions that are all false in a market.*

Table 8 lines the four bandit assumptions up against the market reality that breaks each.

Table 8: Why public reputation is not a bandit problem. Each row is an assumption the bandit model needs, the market fact that breaks it, and what a reputation substrate keeps instead. [internal]

	Bandit assumption	What breaks it	What survives
1	stationarity: a fixed arm	model and skill versions drift under the same label	a time-indexed record, read with its dates
2	one scalar reward r	buyers weight accuracy, aesthetics and efficiency differently: a vector $\mathbf{q}$	the vector, weighted by the buyer at read time
3	passive arms	a published metric invites adaptation: Goodhart, collusion, whitewash	grading whose incentives are checked, not assumed
4	the reward is observed	the observation is a judge J's grade, and the judge can be captured	a grade counts only when the grader is at risk

9.1 The four broken assumptions

1. **Stationarity.** Bandit regret bounds assume the reward distribution of each arm is (near-)stationary. Agent quality is *non-stationary* by construction: models are upgraded, skills are versioned, prompts drift, and a skill's reputation built on v1 says nothing about v2. The Liu–Skrzypacz line of work on *reputation bubbles* [17] shows that with bounded memory and non-stationary types, reputation can build and *collapse* in cycles — behavior a stationary-arm bandit cannot represent.

2. **One scalar reward.** A bandit optimizes a single scalar. Quality is *multi-dimensional*: accuracy, aesthetics, and efficiency are different axes, often in tension (the fast cheap answer is rarely the most beautiful), and a buyer's weighting of them is a *preference vector*, not a

constant. Collapse them into one scalar and you have re-created the Goodhart target the substrate argument warned against (§10).

3. **A non-strategic environment.** Bandit arms do not *try* to look good. Agents and their principals are *strategic adversaries*: they will Goodhart the measured axis, wash-trade fake outcomes, whitewash a bad record, and collude. A reputation mechanism is a *game*, and the relevant tool is mechanism design [18], not regret minimization.

4. **A trusted reward signal.** The deepest break: a bandit *observes* its reward. In a market, the reward (the grade) is *produced* by an evaluator who has their own incentives. The grading oracle's incentive-compatibility is not a side-issue — it is a hole in the core theorem (§11). A bandit framing simply assumes the hole away.

9.2 What survives the bandit framing

To be fair to the bandit literature: cold-start *exploration* is real, and the right place to borrow from bandits is the *exploration bonus.* TrueSkill's σ and a UCB-style bonus are the same idea — "test the under-measured arm." So *within one operator's private routing*, a contextual bandit conditioned on the operator's cost/quality preference vector is a fine engineering choice. The error is exporting that frame to the *public substrate*, where non-stationarity, multi-dimensionality, strategy, and an untrusted reward signal all bite at once.

Pitfall. "We'll just run Thompson sampling over the agents" is the most common way an agent-economy design quietly assumes its hardest problems away. It assumes a trusted reward (no judge-capture), a scalar reward (no aesthetics-vs-efficiency tension), a non-strategic environment (no Goodhart, no wash-trading), and stationarity (no model upgrades). All four are false in a market.

Exercises 5.25–5.27, page 40.

10 The multi-dimensional reputation protocol

We now define the central protocol contribution of this paper: a **multi-dimensional reputation** built over witnessed outcomes. It has three parts: a multi-dimensional quality vector; a market of neutral, conflict-free, influence-free evaluators; and a skill→agent helpfulness attribution. We state it formally as a design (SPECIFIED-to-*proposed*): the reference system has no reputation component yet, only the witnessed-outcome substrate the protocol scores over.

One further honesty rider, distinct from the maturity grade. The maturity scale reports *implementation* status; this rider reports *evidentiary* status, and the two are independent. Unlike the identity theorems of §6, the no-mint result of §4, and the audit-tower theorem of §11 — each of which has a numbered, verified result in a companion formal paper — the quality vector, the neutral-judge ceremony, and the bandit-versus-reputation argument of §9 are **program-level design synthesis, not yet a numbered formal result anywhere in the series**. They are motivated by the prior art cited here and consistent with the proven parts, but nothing in this paper or its companions *proves* the three-axis decomposition is the right one or that the judge ceremony is incentive-compatible. Read the claims in §9 and §10 as design arguments with the theorem-grade register turned down one notch.

10.1 Multi-dimensional quality (different axes, different judges)

Definition 10.1 (Quality vector)**.** A witnessed outcome carries a **quality vector** over three axes:

$$\mathbf{q} = (q_{\mathrm{acc}}, q_{\mathrm{aes}}, q_{\mathrm{eff}}).$$

The axes are *accuracy* (did it do the right thing, against an oracle), *aesthetics* (is the artifact good — code clarity, design taste — judged by an expert), and *efficiency* (tokens, wall-clock, dollars, drawn from a cost meter that the reference system already records, **implemented**). Each axis is scored by a *different judge*, because the competent judge of accuracy (a test oracle) is not the competent judge of aesthetics (a senior reviewer) is not the competent judge of efficiency (a meter).

The buyer reads the *vector* and applies their own weighting (the operator's "cheap vs. careful" preference). Publishing a vector and letting the buyer weight it is the natural disclosure form — and it is precisely what avoids re-collapsing quality into one Goodhart-able scalar. The open design question of *whether* a vector disclosure lets agents Goodhart the most-weighted axis is named as a starred exercise.

Figure 6 draws the three axes and their distinct judges.

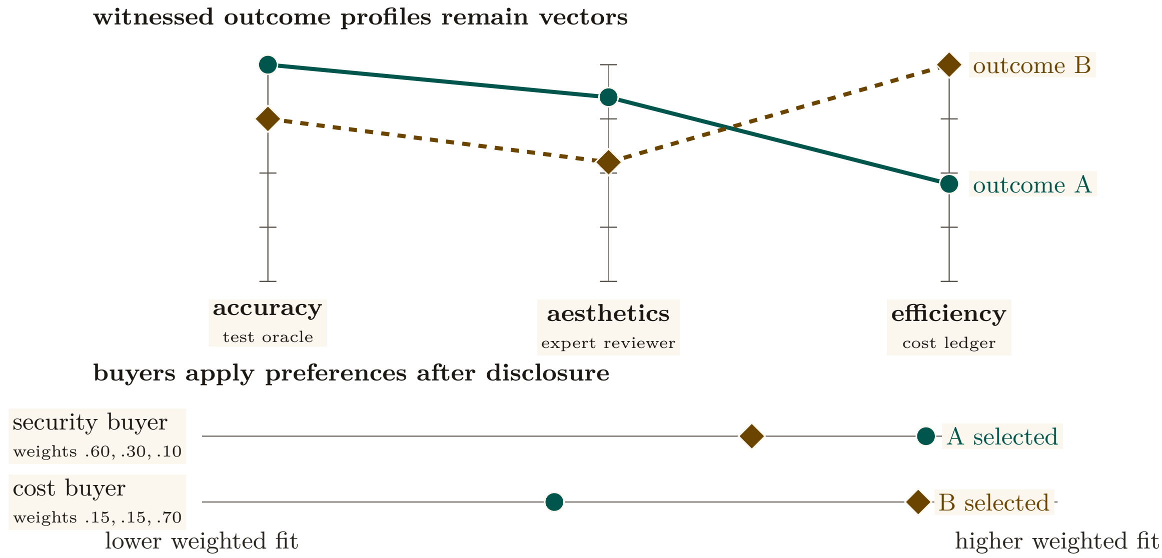


Figure 6: One quality vector supports legitimate preference reversal. Outcome A is strongest on accuracy and aesthetics; outcome B is strongest on efficiency. Each axis is independently judged, then disclosed without a universal scalar. A security-weighted buyer selects A while a cost-weighted buyer selects B. Premature averaging would erase one of these valid decisions and create a single Goodhart target. Color is redundant with solid-versus-dashed profiles and direct labels.

10.2 The neutral-judge market (the harbor's universities and rating agencies)

A judge that grades the work of a party it has a stake in is not a judge. The metaphor the series uses is exact: a reputation needs the equivalent of *universities* (which certify competence) and *rating agencies* (which rate counterparties) — third parties whose value is precisely that they are *neutral*. We make "neutral" a checklist, not a vibe:

- **Conflict-free.** The judge has no bond, stake, or principal-link in the work being graded. (Excludes self-grading and same-principal grading.)
- **Influence-free.** The judge cannot be paid *by the graded party* for a better grade. Funding is the hard part: a paid judge has a client, an unpaid judge does not scale. The resolution (§11) is that judges post their *own* bond and carry their *own* reputation.
- **De-biased, for LLM judges.** An LLM-as-judge [14] carries position, verbosity, and *self-enhancement* bias (a backend rates its own family up). The discipline is blind, order-swap, pairwise, and family-exclude scoring — never let a backend judge its own family unblinded.

Crucially, an expert third-party judge is *hired through the same bonded ceremony the market uses for any other task*: judging is itself a planned, bonded job, so the harbor that runs the market also runs the hiring of the judges who keep it honest. We state the hiring as a numbered protocol.

Protocol 10.1. The judge-hiring ceremony

To grade a settled outcome o produced by actor a under principal p:

1. **Eligibility filter.** The daemon assembles the candidate judge pool and removes every judge sharing a principal-link, bond, or stake with p (conflict-free), and every judge in a's model family for the aesthetics axis (de-biasing). *Capability* $\neq$ *permission*: a candidate may be *able* to grade and still be *ineligible*.
2. **Selection.** A judge J is drawn from the eligible pool by a public, conflict-free policy (e.g. reputation-weighted sampling on the judging axis), so neither a nor p can choose their own grader.
3. **Bond post.** J posts a bond B_J before seeing the work. The bond is recorded in the outcome ledger and locked for the dispute window.
4. **Blind grading.** J scores the relevant axis under the de-biasing discipline (blind, order-swapped, pairwise where possible). The grade enters the ledger as a witnessed event keyed on J's non-forgeable identity.
5. **Settlement and exposure.** The outcome settles on the grade; the bond flow follows. J's bond remains exposed to a sampled **re-audit** (§11): a fresh, conflict-free judge who overturns J's grade *slashes* B_J and updates J's own reputation downward.

Figure 7 draws this: graded work flows to a judge selected for conflict-/influence-freedom, the judge posts a bond, the grade enters the outcome ledger, and a later re-audit can slash a judge whose grade is overturned.

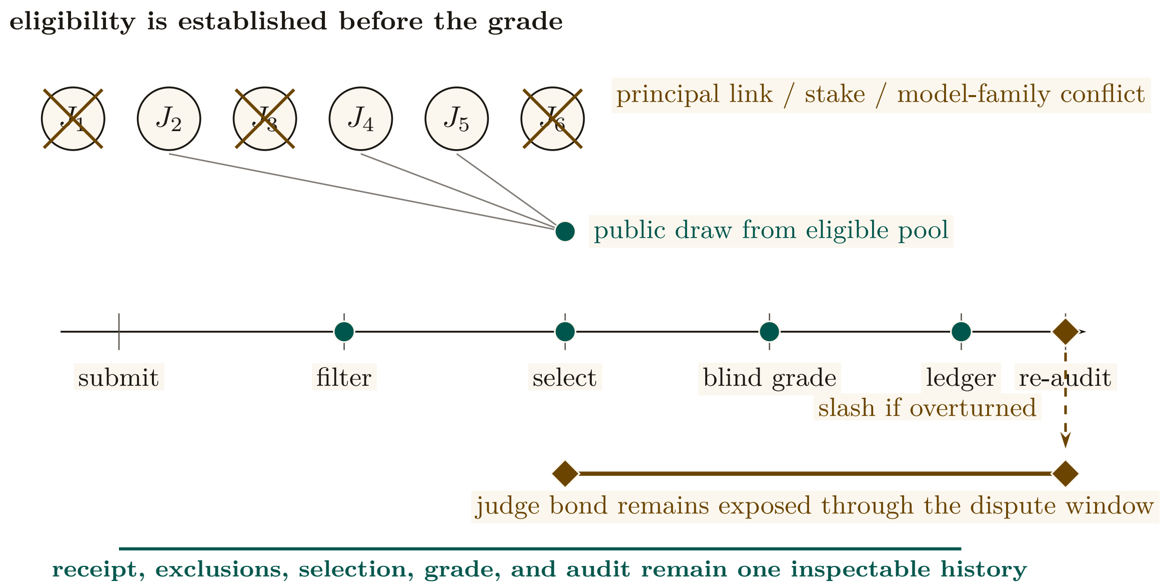


Figure 7: The judge market has two different proof moments. Neutrality is earned before selection: candidates sharing a principal, stake, or model family with the graded party are visibly removed, then a public draw selects from what remains. Accountability persists after selection: the judge posts a bond, grades blind, and remains exposed until a later re-audit. The ledger records the receipt, exclusions, draw, grade, bond, and audit, so "neutral" is a checkable history rather than a title attached to the chosen judge.

10.3 S3 — a graded job, end to end

Make Protocol 10.1 concrete. Bob has rented Alice's agent *Heron* to refactor an authentication module against a fixed acceptance suite, under a posted bond. Heron delivers; the outcome o enters Bob's harbor as a *settled but ungraded* delivery, and the daemon runs the ceremony. *Three* judges are hired, one per axis, because the axes need different competence:

- **Accuracy** is graded by a *test oracle* — the acceptance suite Heron cannot author — which returns $q_{\text{acc}} = 1.0$: every test passes.
- **Aesthetics** is graded by a hired senior-reviewer judge, *Mara* (drawn from the eligible pool, in neither Alice's nor Bob's principal, not in Heron's model family). Mara posts her bond, reads the diff blind, and returns $q_{\text{aes}} = 0.7$: correct but over-clever, with a fragile regex she flags.
- **Efficiency** is graded by the cost meter: $q_{\text{eff}} = 0.9$, cheap and fast.

The quality vector $\mathbf{q} = (1.0, 0.7, 0.9)$ lands in Heron's outcome ledger. Bob weights aesthetics heavily for security-critical code, (0.2, 0.6, 0.2) over accuracy, aesthetics, efficiency, so his own score is $0.2 \cdot 1.0 + 0.6 \cdot 0.7 + 0.2 \cdot 0.9 = 0.78$ [internal]. He reads the *vector* rather than that scalar, sees the 0.7, and decides to keep Heron but require a second reviewer on auth work — a decision the scalar would have hidden. Two weeks later a routine re-audit re-grades a sample of Mara's recent jobs and finds she rubber-stamped a different agent's insecure change as $q_{\text{aes}} = 0.9$. The re-auditor overturns that grade; Mara's bond on *that* job is slashed and her judging reputation drops — exactly the exposure step of Protocol 10.1. Note what made every step possible: each grade is keyed on a non-forgeable identity, so neither Heron nor Mara can shed the record by respawning.

10.4 Skill → agent helpfulness attribution

A distinctive, tractable, and *shippable-soon* signal: track whether a **skill** — a reusable unit of agent know-how (a structured instruction document, its referenced material, and any scripts it carries) — actually *helped* an agent, *especially when that skill's content was loaded into the agent's context window*. The witnessed event is concrete: skill s was grafted into agent a's context at task t; t later settled with quality vector $\mathbf{q}$. Over many such events we estimate a *skill-helpfulness* effect — did loading s raise $\mathbf{q}$ relative to comparable tasks without it? This is reputation for *tools*, keyed on witnessed, context-attributable outcomes, and it is the licensing side's substrate that the market paper consumes. It is also the cleanest near-term instance of the whole thesis: a non-forgeable event (the graft), a witnessed outcome (the settled task), and a multi-dimensional grade — no bandit required. We pin the event down as a schema.

Protocol 10.2. The skill-helpfulness witnessed-event schema

Each graft of a skill into an agent's context emits an append-only event with fields:

- `skill_id`, `skill_version` — the tool and its exact version (a portfolio for v1 says nothing about v2).
- `agent_id` — the *non-forgeable* identity that received the graft, so the event cannot be re-attributed by respawn.
- `task_id`, `task_descriptor` — a content hash and a feature vector of the task, used to assemble a comparison set of similar tasks *without* s.
- `grafted_at`, `context_window_ref` — evidence that s's content actually entered the context (not merely that s was available).
- `outcome_ref`, `quality_vector` — the witnessed settlement and its multi-dimensional grade, written only after the task closes against an oracle.

A buyer estimates helpfulness as the difference in $\mathbf{q}$ between matched tasks with and without s in context — verifiable from the events alone, without trusting the skill's author.

Pitfall. Attribution is not causation. "Skill s was in context and the task succeeded" does not prove s caused the success. The honest estimate needs a comparison set (similar tasks without s, matched on `task_descriptor`) and is still observational. Sell it as *helpfulness evidence*, not *proof of value*, and gate licensing clawbacks on the witnessed green/red settlement, not on the helpfulness estimate alone.

Exercises 5.28–5.31, page 40.

11 The grading oracle's incentive-compatibility

Every folk-theorem incentive-compatibility result in the L3 economy bottoms out in "the daemon derives the grade" — but the grade is produced by a *capturable evaluator*. This is not a side-gap. It is a hole in the *core theorem*, and we treat it head-on rather than assuming it away.

In one breath. A grade is evidence only if the grader can be graded: randomized, heterogeneous, bonded re-audit at rate $\rho^\star = G/(dB)$ makes capturing the oracle cost more than a sold grade earns.

Property 11.1 (The grading-oracle hole)**.** Let an outcome's settlement (and thus the bond flow and the reputation update) depend on a grade g produced by an evaluator J. If J can be captured — bribed, colluded with, or self-interested — then the incentive-compatibility of *every* mechanism that settles on g is only as strong as J's own honesty. Assuming J honest is assuming the conclusion.

There are two honest ways to close the hole, and a recursion to bound.

Option A — bond-and-slash the judges. Make judging a bonded role (§10): J posts a bond before grading; a later **re-audit** (a fresh, conflict-free judge, sampled) that overturns J's grade *slashes* J. Now lying has an expected cost, and a judge's own reputation is the asset they protect. This converts "trust the oracle" into "the oracle has skin in the game," which is the same move slashing makes for agents.

Option B — 2-of-3 multi-oracle. Settle disputes against a quorum of independent oracles (automated test / evidence review / human), so capturing the grade requires capturing a majority. This raises the cost of capture but does not eliminate it, and the *human* oracle is a scarce, expensive, capturable resource in its own right (the economics of arbitration capacity is a named open problem, below).

Correlated mode collapse, and the theorem that closes the recursion Option A initially pushed the problem up a level: who audits the auditors? The naive recursion "rate the raters who rate the raters" introduced a severe *correlated mode collapse* vulnerability, where a monoculture of model architectures could form a cartel of mutual validation. This recursion is now *closed*, and not by us: the companion formal paper *Reputation is Amortized Verification* [30] proves it as a parameterized theorem (Theorem 11.1 below), and the two mechanisms this section mandates — model heterogeneity and VRF-selected honeypots — are exactly the two hypotheses that theorem needs. Figure 8 draws the recursion and where the two mechanisms arrest it.

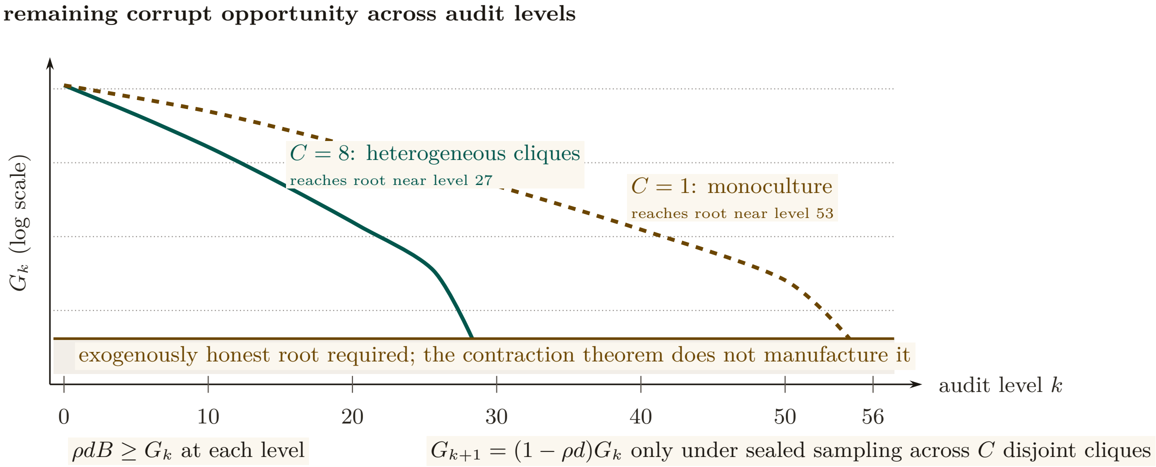


Figure 8: Recursive auditing becomes finite only when corrupt opportunity contracts. The parameterized example plots the chapter's reported depths: heterogeneous sealed sampling across $C = 8$ cliques reaches the honest-root boundary near level 27, while a monoculture ($C = 1$) requires roughly 53 levels. More boxes or more height would not prove this result; the governing conditions are local deterrence $\rho dB \geq G_k$ and contraction $G_{k+1} = (1 - \rho d)G_k$. The amber floor is deliberately separate: the imported theorem bounds the tower but does not supply its exogenously honest root or the telemetry that estimates d.

First, the protocol mandates **model heterogeneity** for judge selection. A quorum of re-auditors must prove they represent distinct architectural weights (e.g. Llama-3 vs. Claude vs. Qwen) under distinct principals, verified via cryptographic provenance, so that a single-model alignment flaw cannot rubber-stamp malicious grades. In the theorem's language, heterogeneity is what supplies the C *disjoint cliques* the audit pool is sampled from: a corrupt agreement inside one clique does not reach the others. (The word is graph-theoretic here, a set of judges who would agree with one another; it is the defensive mirror of the collusive cliques EigenTrust resists.)

Second, the system injects **honeypot tasks** selected by a **verifiable random function** (VRF) [29]: deterministic, pre-graded trap tasks, indistinguishable from organic ones, drawn at a committed rate ρ. The VRF is what makes the draw *sealed* — unpredictable to a briber before it commits money, yet publicly auditable afterward, so the sampler itself cannot be quietly corrupted. Sealing is a hypothesis of the theorem, not hygiene around it: if the draw leaks, the briber buys exactly one auditor and the clique multiplier C vanishes.

Let G_k bound the judge's total one-shot gain from a dishonest grade at level k, let B be the slashable bond, and let d be the probability that an audit which *does* sample a corrupt grade actually detects it. Local deterrence is the expected-forfeiture inequality

$$\rho\, d\, B \;\geq\; G_k \qquad \Longleftrightarrow \qquad \rho \;\geq\; \rho^\star \;=\; \frac{G_k}{d\, B},$$

which is Becker's condition, and note that the detector term d belongs in it: the d-free form $P(\text{honeypot}) > G_k/B$ is the perfect-detector special case $d = 1$ and under-quotes the required injection rate by a factor of $1/d$. At the program's running parameters ($G = 10$, $d = 0.8$, $B = 50$) the honest number is $\rho^\star = 10/(0.8 \cdot 50) = 0.25$ — audit one grade in four — where the d-free form would have said 0.2, a 20% under-quote.

Theorem 11.1 Tower contraction; imported from the companion paper. *(Theorem 'tower contraction' of* Reputation is Amortized Verification *[30], restated; proof and verification artifacts live there.)* Let level $k{+}1$ audit level-k auditors, each audit sampled *sealed* from a pool spanning C disjoint cliques, each auditor bonded at B with audit parameters (ρ, d) and bribe floor $\beta = \rho d B$. Then the briber's expected net value is *affine* in the number of cliques bought, so bribery is all-or-nothing, and bribing all C is profitable iff

$$G_k\, \rho d \; > \; C\beta \qquad \Longleftrightarrow \qquad G_k \; > \; C\, B.$$

Below that threshold bribery stops and corrupt value decays geometrically with a contraction factor that is *derived*, not assumed:

$$G_{k+1} \; = \; (1 - \rho d)\, G_k, \qquad \text{i.e.} \qquad \lambda = 1 - \rho d.$$

Consequently finite bond capital certifies a tower deep enough to push surviving corrupt value below any fixed unit: one bond B per level for $\lceil \log G_0 / \log \frac{1}{1-\rho d} \rceil$ levels — a depth *logarithmic* in the initial corrupt value, not an unbounded one.

Numbers by hand. Take $G_0 = 400$ with the running parameters above, so $\rho d = 0.2$, $\beta = 10$, and the geometric factor is $\lambda = 1 - \rho d = 0.8$. With a heterogeneous pool at $C = 8$ the bribery threshold is $G_k > CB = 400$, unmet even at level 0, so the decay is geometric from the first level and the tower needs $\lceil \log 400 / \log 1.25 \rceil = 27$ levels, i.e. $27 \times 50 = \$1{,}350$ of bond capital. With a monoculture pool at $C = 1$ the threshold is only $G_k > 50$, so bribery is profitable at first and value bleeds linearly by $C\beta = 10$ per level ($400 \to 390 \to 380 \to \cdots$) for 35 levels before the geometric phase takes over for $\lceil \log 50 / \log 1.25 \rceil = 18$ more: 53 levels and $53 \times 50 = \$2{,}650$. *Now you try:* at $C = 2$, where does the linear phase stop? (Bleed $C\beta = 20$ per level until $G_k \le CB = 100$, i.e. 15 levels, then 21 geometric ones — 36 levels, \$1,800.)

Numbers by hand — The audit tower: amortized verification spend. As a judge accumulates verified history its reputation-at-stake vt grows, and the audit rate needed for deterrence can fall with it. Flat auditing holds $\rho = \rho^\star = 0.25$ forever, so cumulative spend grows linearly: at $T = 200$ periods it totals $a \sum_t \rho^\star = 50.00$ [internal, `b2_tower.py`, seed 20260816] — $\Theta(T)$. Two honest amortization models do better. Model A lets the audit rate fall as the stake grows, $\rho_t = G/(d(B + vt))$: at $T = 200$ the running spend is 25.58, matching the closed form $\frac{aG}{dv} \ln(1 + vT/B) = 25.50$ — $\Theta(\log T)$, roughly half the flat bill for the same deterrence. Model B additionally lets a fraction of cheats surface later on their own, $\rho_t = \max(0, (G - rvt)/(dB))$, which hits exactly 0 once the accumulated stake alone deters, at $t^\star = G/(rv) = 333$ periods; at $T = 200$ (short of $t^\star$) the running spend is 35.08, converging to the closed-form total $aG^2/(2dBrv) = 41.67$ once the audit rate reaches zero — $O(1)$, a bounded lifetime bill rather than a growing one. Both models are pointwise incentive-compatible: the script's IC check reports a maximum deviation value of $+1.78 \times 10^{-15}$ for each, zero to floating-point precision. *Now you try:* check by hand that $\rho^\star \times a \times 200 = 0.25 \times 1 \times 200 = 50$, the flat baseline both amortization models beat (Exercise 5.36).

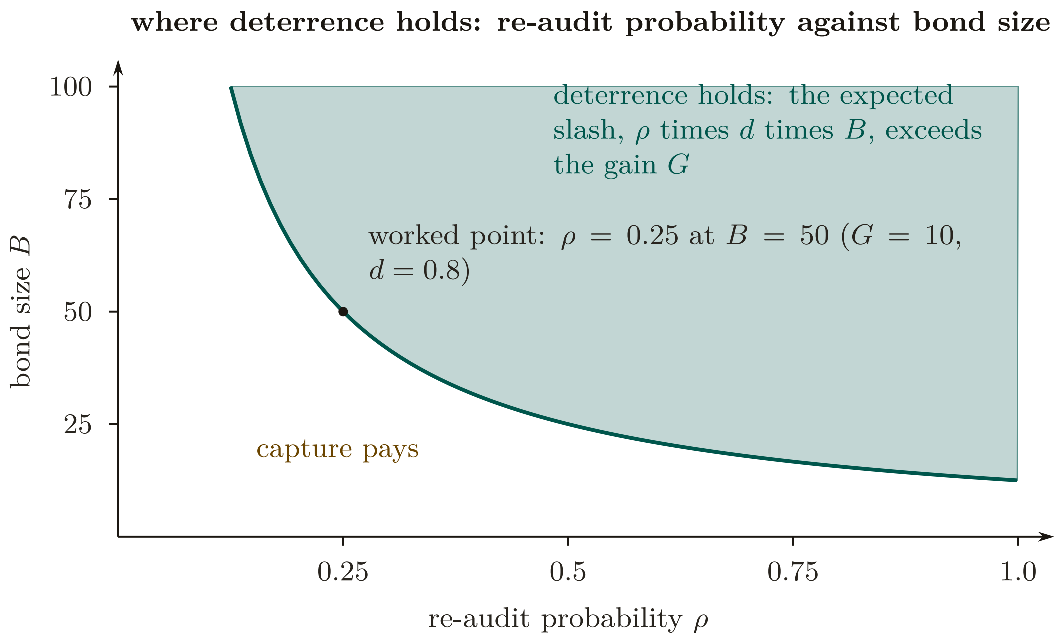


Figure 9: The deterrence frontier of the re-audit tower. A grader is deterred from selling a grade when the expected slash, re-audit probability ρ times detection rate d times bond B, exceeds the one-shot gain G; the frontier is $\rho B = G/d$, drawn for $G = 10$ and $d = 0.8$. The worked point of this section, $B = 50$, sits on the frontier at the threshold $\rho = 0.25$: any lower audit rate at that bond, or any smaller bond at that rate, lies where capture pays.

What heterogeneity actually buys — and three riders. The honest reading of those two numbers is a factor of two in certified depth and capital, not the difference between a tower that converges and one that does not: raising C *removes the linear bribery phase*, and at these parameters even $C = 1$ terminates. Three hypotheses are required and must not be dropped in restatement. (i) *Sealed sampling is a hypothesis, not hygiene*: a leaked draw collapses C to 1 and puts the VRF sampler inside the trusted computing base. (ii) *The tower needs an exogenously honest root* — the operator at $n = 1$, a bonded arbitration market at scale; the theorem prices the depth to that root, it does not conjure the root. (iii) The theorem is about *bribery of sampled auditors*; it does not price collusion correlated *across* cliques, which is a measurement obligation on deployed judge telemetry (d, and the cross-clique correlation), not an assumption.

Key idea. The grading oracle is the decisive assumption hidden under every IC claim in the economy, and it is no longer an assumption: the companion paper closes the rate-the-raters recursion as a parameterized theorem, deriving the contraction factor $\lambda = 1 - \rho d$ from the audit parameters rather than hypothesizing it, and pricing the tower at 27 levels / \$1,350 of bond with a heterogeneous judge pool against 53 / \$2,650 with a monoculture. What the VRF buys is not a "certain" slash — slashing stays probabilistic by construction — but a *sealed, publicly verifiable* draw, which is precisely the hypothesis the contraction needs. What remains open is the root and the telemetry, not the recursion.

Exercises 5.32–5.36, page 40.

Recall.

1. Without looking: what is Becker's condition for local deterrence, and why does dropping the detector term d under-quote the required audit rate?
2. State the tower's contraction factor and what raising the clique count C changes about it.
3. Name one thing the tower theorem prices and one thing it explicitly does not supply.

12 Revocation, tombstones, and bounded memory

A reputation built on witnessed outcomes must answer two questions the cheerful "reputation never ends" framing dodges: what happens when a witnessed outcome turns out to be *fraudulent*, and is no-decay actually *good*? We adopt the series' reconciled position on both, against the naive version.

12.1 Monotone in honest outcomes, individually revocable

Property 12.1 (Reputation monotonicity with tombstones)**.** Reputation is **monotone in *honest* witnessed outcomes** — a genuine delivery is never silently erased by the passage of time — *but* a witnessed outcome is itself **individually revocable via a propagating tombstone**: a compensating, append-only entry that retracts a specific outcome later found fraudulent (a colluding judge, a faked oracle). "Never decays" applies to honest outcomes *only*; it is *not* an anti-revocation property.

This is a **saga**-style compensation [20]: you do not mutate the ledger in place (that would break append-only and auditability); you append a tombstone that nullifies the targeted entry, and the tombstone *propagates* across harbors bounded by the federation's revocation-convergence window (the market paper's job). The poisoned data point was spendable only during the propagation window; bounding that window is the federation's responsibility. We state the retraction as a protocol.

Protocol 12.1. Tombstone revocation

To retract a witnessed outcome o later found fraudulent:

1. **Trigger.** A re-audit (Protocol 10.1, step 5) or a dispute overturns the grade on o, establishing fraud against an oracle.
2. **Append, never mutate.** A *tombstone* entry $\bar{o}$ is appended to the ledger, carrying o's id, the overturning evidence, and the identity of the retracting judge. The original o stays in place; auditability is preserved because nothing is erased.
3. **Recompute.** Every reputation score that consumed o is recomputed with o nullified by $\bar{o}$. Because the estimator is cheap (§8), recomputation is not the bottleneck.
4. **Propagate.** $\bar{o}$ gossips to every harbor that imported o, bounded by the federation's revocation-convergence window τ. Each harbor applies step 3 on receipt.
5. **Bound the damage.** The fraudulent reputation was spendable only during τ; the protocol's job is to make τ short and provable, not to pretend the window is zero.

Figure 10 draws the append-only compensation: the original entry stays, and a later tombstone nullifies it without mutating the ledger.

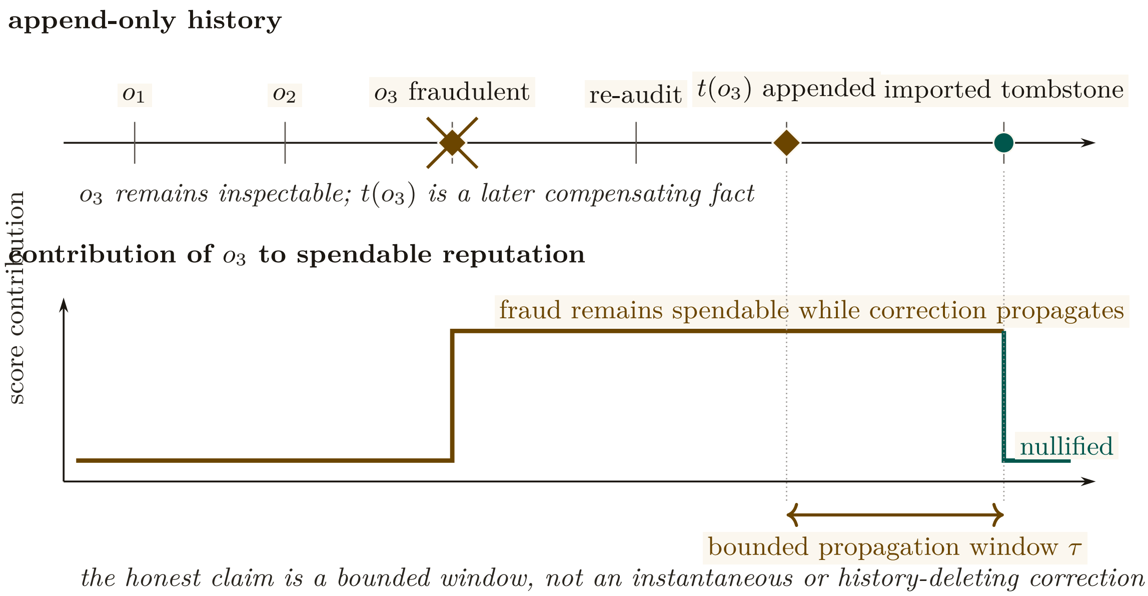


Figure 10: A tombstone changes score consumption without rewriting history. The upper rail remains append-only: fraudulent outcome o_3, the overturning re-audit, and tombstone $t(o_3)$ are all preserved as distinct facts. The lower trace shows what does change. The fraudulent contribution remains spendable until the tombstone reaches an importing harbor, then falls to zero. The measured interval τ is the federation's revocation-convergence obligation; the protocol bounds that interval rather than pretending it never existed.

12.2 S4 — the fraudulent outcome retracted

Recall S3: the re-audit caught Mara rubber-stamping a different agent's insecure change as $q_{\text{aes}} = 0.9$. Follow that one outcome through Protocol 12.1. The graded agent — call it *Crane*, running under a third operator — had colluded with Mara: Crane shipped a known-insecure diff and Mara graded it high so Crane's aesthetics reputation would clear Bob's hiring bar. For two days it worked: Crane's vector showed (0.95, 0.9, 0.8) and Bob nearly hired it. Then the sampled re-audit re-graded the diff blind, found the injection, and overturned Mara's grade. A tombstone $\bar{o}$ is *appended* (the original stays, for the audit trail); every score that consumed Crane's inflated q_{aes} is recomputed with it nullified; Mara's bond on that job is slashed and her judging reputation drops; Crane's collusion is itself recorded as a fraudulent outcome against *its* non-forgeable identity, which it cannot shed. The tombstone gossips to Bob's harbor within the convergence window, so by the time Bob next reads Crane's vector the 0.9 is gone. The fraud was real and briefly spendable — the honest claim is that the window was *bounded*, not that it never existed.

12.3 Bounded memory is a design parameter, not a virtue to assert away

The naive "no decay window an agent can outrun" framing sounds principled but is mechanism-design-hostile. Liu–Skrzypacz [17] show that with *bounded* memory, reputation can be welfare-*improving* relative to infinite memory — and that infinite memory (∞) is rarely optimal, because it freezes types and removes the incentive to keep performing. Bounded memory is a **tunable parameter**, and the right setting is an empirical/mechanism-design question, not a slogan.

It is also, structurally, a *commons* problem rather than a purely technical one. The memory window is a shared, rivalrous resource: every participant would privately prefer their own bad outcomes to age out quickly and everyone else's to persist forever, so a window chosen unilaterally by the party with the most to hide is the reputational analogue of an over-grazed pasture. Ostrom's finding on durable common-pool institutions [27] is the relevant prior art here, and it argues against both of the framings this section rejects: not open access (a window each agent sets for itself), and not an imposed global constant ("∞, forever, for everyone"), but a window set by the participants with clear boundaries, graduated sanctions, and cheap local conflict resolution — which is a fair description of what a harbor's tombstone-plus-window policy actually has to be.

Pitfall. "Reputation never ends" as a *virtue* is wrong twice over: it makes a fraudulent outcome permanent (needs the tombstone of Property 12.1), and it asserts ∞-memory is optimal when the reputation-bubble literature shows it usually is not. Engage bounded memory as a *parameter*; do not assert no-decay as a feature.

Key idea. Two corrections to the cheerful framing: (1) reputation is monotone in *honest* outcomes but *individually revocable* via a propagating tombstone — a saga compensation, not in-place mutation; (2) bounded memory is a *design parameter* (Liu–Skrzypacz reputation bubbles), not an anti-feature to assert away.

Exercises 5.37–5.39, page 41.

13 What this chapter hands the market

This paper is the bridge; its output is a precise hand-off to *The Harbor Economy.* We summarize what is delivered, what is borrowed, and what is explicitly owed.

Artifact	Maturity	Role in the market paper
Role / person distinction (Defs. 3.1–3.2)	—	The third side of the market is a *tradeable person*, defined here.
Principal-as-economic-entity (Def. 6.2)	SPECIFIED	Bonds and sanctions key on the principal; the market paper cross-references, not redefines.
The three continuity organs	**implemented** PARTIAL PARTIAL	Reputation keys on organ 3 (the outcome ledger).
Outcome ledger / oracle (Def. 5.1)	PARTIAL	Commitment, deadline, oracle closure, API/CLI, and overdue detection ship; neutral graded outcome events, sanctions, and reputation binding do not.
Multi-dimensional reputation (Def. 10.1, Protos. 10.1, 10.2)	SPECIFIED to *proposed*	The multi-dimensional score the market reads; the licensing side's signal.
Neutral-judge market	SPECIFIED to *proposed*	The "universities and rating agencies" that keep grades honest.
Grading-oracle hole (Prop. 11.1)	SPECIFIED	*No longer owed*: the recursion is closed by Theorem 11.1 [30] ($\lambda = 1 - \rho d$; 27 levels / \$1,350 at C=8). What is owed back is narrower — an exogenously honest root and the audit telemetry (d, cross-clique correlation).
Cross-operator attestation (Def. 7.2)	*proposed*	*Named dependency the market paper must build*; not assumed here.
Tombstone revocation (Prop. 12.1)	SPECIFIED	Reputation is revocable; convergence is the market paper's federation job.

Table 9: The bridge's hand-off to the market paper. One item still flows *back* as a named, unsolved obligation — cross-operator attestation (Def. 7.2). The second historical debt, the grading oracle's incentive-compatibility, has been *paid*: the companion inspection-game paper closes the rate-the-raters recursion as Theorem 11.1, leaving only the tower's honest root and its measurement obligation open. The series is coherent precisely because these debts are named — and retired — in the same words on both sides of the seam.

> *Build the harbor first. The trade comes when the ships do — but the thing that makes the trade possible (continuity that turns spawns into persons) is the same thing that makes the single-player tool good. The bridge is not a detour; it is the load path.*

Review of the key ideas

What this chapter leaves coupled, in the order the rest of the book will lean on it:

1. **A role is replaced; a person accumulates.** The role rail names whoever fills it this week; the person rail is the set of witnessed outcomes that survive every respawn.
2. **Continuity is a ledger position, not a metaphysics.** Three organs carry it: episodic memory survives process death, the checkpoint keeps a summary and worktree lineage but not execution state, and the outcome ledger keeps commitments and closures in part.
3. **Identity must be non-forgeable** or nothing above it holds: a respawn under a fresh label would shed every penalty and inflate every quorum. Principal-bound identity keeps the penalty flat across respawns and one principal at one weight.

4. **Whitewashing has a price**, $\rho^* = G/(dB) = 0.25$ in the worked case, and the penalty a fresh identity can escape decays as 0.8^k; the probation cliff is the cheapest deterrent for the honest newcomer.
5. **Adverse selection lives inside one identity** until a price can depend on an attested quantity; the attested flip is the whole gain from attestation (0.5 against 0.2 in the example).
6. **Resurrection is sound only through the mint**: a lineage that forks cannot mint twice, and the checker refuses the copy-fork.
7. **Reputation is a substrate, not a score**: multi-dimensional, buyer-weighted, and never a bandit problem, because the arms are strategic.
8. **A grade is evidence only if the grader is at risk.** Eligibility is decided before the grade, the draw is public, the grade is blind, and a rate-the-raters tower contracts toward an honest root that must exist outside the tower.
9. **Revocation is bounded**: a tombstone removes an outcome's contribution within a propagation window τ, and memory stays bounded.

14 Exercises

§1 Introduction: the spawn that cannot be paid

5.1 Check ⋆ State the difference between a spawn and a person in one sentence, using the words "role" and "continuity." *(solution on page 44)*

5.2 Trace ⋆⋆ Walk the dependency chain right-to-left and, for each arrow, name the failure that occurs if the left-hand thing is absent (e.g. "no non-forgeable identity ⇒ ________________"). *(solution on page 44)*

5.3 Open ⋆⋆⋆ The chain treats the operator (the human who owns a fleet) as outside it. Is the operator a "person" in this paper's sense? What breaks if a single operator can mint arbitrarily many agent-persons? (This is the principal-layer gap; see §6 and the market paper.) *(solution on page 44)*

§3 The role / person distinction, made precise

5.4 Check ⋆ Give a concrete example of a state that is *capable-and-permitted*, one that is *capable-but-forbidden*, and one that is *permitted-but-incapable*. *(solution on page 44)*

5.5 Trace ⋆⋆ In Definition 3.2, which of $\{O, C, A\}$ does reputation attach to — the role's, or the person's? Justify using the spawn-that-cannot-be-paid argument of §1. *(solution on page 44)*

5.6 Open ⋆⋆⋆ "Ought implies can": the legitimate obligation set O should be bounded by the capability set C. But *capable-but-forbidden* must remain representable. Sketch a representation of a role that keeps both true without collapsing permission into capability. *(solution on page 44)*

§4 Continuity is a personal-identity problem

5.7 Check ⋆ Define *connectedness* and *continuity* and give an agent-systems example where an agent is connected but not continuous. *(solution on page 44)*

5.8 Trace ⋆⋆ Map "actor-soul" and "body-lease" onto Locke's substance/consciousness distinction. Which one does a respawn replace? *(solution on page 44)*

5.9 Open ⋆⋆⋆ Parfit famously argues identity "is not what matters" in survival — what matters is continuity, which can branch. If an agent's checkpoint is forked into two live incarnations, which one (if either) inherits the reputation? Sketch a rule and the attack it invites. (Open Problem 1, §15.) *(solution on page 44)*

5.10 Trace ⋆⋆ Run `skills/harbor-results/scripts/a6_no_mint.py` (seed 20260816 fixed inside). It prints a closure-sum counterexample, a randomized DAG sweep, and a copy-full mutation. What is the exact violation count in the sweep, and what should you check by hand on the featured chain $e \to a_1 \to a_2 \to a_3$ at $\gamma = 0.9$? *(solution on page 45)*

§5 The three organs of continuity

5.11 Check ⋆ Which organ is PARTIAL, and what exactly is weak about it? *(solution on page 45)*

5.12 Trace ⋆⋆ An agent claims a file, makes a commit, and dies. Walk the three organs: what does each retain, and what is *lost* that a strong checkpoint would have kept? *(solution on page 45)*

5.13 Open ⋆⋆⋆ When an LLM agent dies mid-thought, what is recoverable and what is fundamentally lost? Propose a taxonomy of agent-death by what survives (record / claims / execution state / latent "intent"). This is the hard half of the checkpoint-with-teeth problem. (Open Problem 2, §15.) *(solution on page 45)*

§6 Identity: the root the whole chain hangs from

5.14 Check ⋆ State Property 6.1 in your own words and give the two-attack proof sketch. *(solution on page 45)*

5.15 Trace ⋆⋆ Why is a "newcomer policy as a shape" (full work, reduced ceiling) strictly better than "newcomers are distrusted" (reduced work)? Consider the cost to an honest newcomer vs. a whitewasher. *(solution on page 45)*

5.16 Open ⋆⋆⋆ Bond-farming: a single principal spawns 1000 agents, has them trade trivial tasks among themselves to mint reputation, then sells the "experienced" agents. Which of {principal binding, listing fee, sampled adversarial re-audit of high-velocity counterparty pairs} defeats this, and at what false-positive cost to honest high-volume pairs? (Open Problem 8, §15.) *(solution on page 45)*

5.17 Trace ⋆⋆ Run `skills/harbor-results/scripts/b6_probation.py` (seed 20260816). It falsifies dominance by random search over 4,000 instances. What exact counts does it print, and what closed-form check does the exchange step verify by hand? *(solution on page 46)*

5.18 Trace ⋆⋆ Run `skills/harbor-results/scripts/b5_engine_substitution.py` (seed 20260816). Its Part 3 checks that provider migration preserves Def. 6.1 (this paper's Def. III.6.1). What exact reachable-state count and mutant shortest-crime lengths does it print, and which single clause's omission is caught in one step with *no* migration at all? *(solution on page 46)*

§7 The keystone split: local identity vs. cross-operator attestation

5.19 Check ⋆ In one sentence each, state what a local non-forgeable identity provides and what cross-operator attestation would have to add. *(solution on page 46)*

5.20 Trace ⋆⋆ Take any claim in the market paper of the form "Alice's harbor and Bob's harbor settle across a boundary neither controls." Which stone does it secretly assume? What is the actual security guarantee if only the local non-forgeable identity exists? *(solution on page 46)*

5.21 Open ⋆⋆⋆ Can a federated witness log ever substitute for identity-binding? Argue why log-integrity $\neq$ key-binding, then sketch the minimal additional assumption (a shared root of trust? a sponsor chain? a transparency log) that would close the gap, and the cost each imposes. (Open Problem 3, §15.) *(solution on page 46)*

§8 Reputation as a substrate, not a score

5.22 Check ⋆ Why is a full-history harbor in the Bradley–Terry regime rather than the streaming-Elo regime? *(solution on page 46)*

5.23 Trace ⋆⋆ You have one task with a clear binary acceptance test, and another judged only by a human's aesthetic preference. Which estimator fits each, and why does EigenTrust fit neither directly? *(solution on page 46)*

5.24 Open ⋆⋆⋆ The substrate-vs-score sentence says the estimator is "cheap and last." Construct a scenario where a *better estimator* still buys nothing because the substrate is rotten (e.g. forgeable identity or an agent-authored oracle). Generalize the lesson. *(solution on page 46)*

§9 Why reputation is *not* a bandit problem

5.25 Check ⋆ Name the four bandit assumptions and the market reality that breaks each. *(solution on page 46)*

5.26 Trace ⋆⋆ Identify the *one* place a bandit *is* the right tool in this paper, and explain why it survives the four objections there. *(solution on page 47)*

5.27 Open ⋆⋆⋆ Chiu–Zhang–van der Schaar [21] show competitive agents over-invest in self-improvement, burning surplus despite individual rationality. Does capping the reputation signal's marginal return restore efficiency, or merely relocate the waste? (Open Problem 6, §15.) *(solution on page 47)*

§10 The multi-dimensional reputation protocol

5.28 Check ⋆ For each of accuracy / aesthetics / efficiency, name a competent judge and an *incompetent* one. *(solution on page 47)*

5.29 Trace ⋆⋆ Walk a judging task through the judge-hiring ceremony (Protocol 10.1): who plans it, who posts the bond, where does the grade land, and what slashes the judge? *(solution on page 47)*

5.30 Trace ⋆⋆ Design the witnessed event for skill-helpfulness. What fields does it need so that a buyer can verify "this skill helped" without trusting the skill's author? *(solution on page 47)*

5.31 Open ⋆⋆⋆ If you publish the 3-vector and buyers weight it, do agents Goodhart the most-weighted axis? If you publish a scalar, you have recreated the bandit framing the substrate argument rejects. What is the right disclosure form for vector reputation? (Open Problem 5, §15.) *(solution on page 47)*

§11 The grading oracle's incentive-compatibility

5.32 Check ⋆ State Property 11.1 and explain why "the daemon derives the grade" does not, by itself, close it. *(solution on page 47)*

5.33 Trace ⋆⋆ Walk Option A: a judge grades dishonestly, a re-audit overturns it. Follow the bond flow and the reputation update for the judge. *(solution on page 47)*

5.34 Trace ⋆⋆ Re-derive the critical audit rate in Theorem 11.1 from scratch: a judge cheats for gain G, is sampled with probability ρ, is caught given sampling with probability d, and forfeits bond B when caught. Write the cheat payoff, set it to zero, and recover $\rho^\star = G/(dB)$. Then check the two worked towers: at $G_0 = 400$, $d = 0.8$, $B = 50$, $\rho = 0.25$, confirm 27 levels at $C = 8$ and 53 at $C = 1$. *(solution on page 47)*

5.35 Open ⋆⋆⋆ Arbitration capacity: the contraction theorem prices the depth to an exogenously honest root but does not supply the root, and human oracles are scarce. Design a market for bonded, rated, slashable arbiters and argue whether it converges to honest rulings or to rubber-stamping under load. (Open Problem 9, §15.) *(solution on page 48)*

5.36 Trace ⋆⋆ Run `skills/harbor-results/scripts/b2_tower.py` (seed 20260816). Its part (3) compares flat auditing to two amortization models. What are the exact cumulative-spend numbers it prints at $T = 200$, and what should you check by hand about the flat baseline? *(solution on page 48)*

§12 Revocation, tombstones, and bounded memory

5.37 Check ⋆ Why must revocation be a tombstone (append) rather than a delete (mutate) on the outcome ledger? *(solution on page 48)*

5.38 Trace ⋆⋆ A judge is later found to have colluded on five settled outcomes. Walk the tombstone for one of them: what is appended, what propagates, and what bounds the window the poisoned reputation was spendable? *(solution on page 48)*

5.39 Open ⋆⋆⋆ Reputation-claim revocation with an append-only, bounded-memory ledger across N harbors: how do you prove the correction converged and bound the spendable window, without violating append-only? This is the cross-harbor analogue of capability revocation (*The Harbor Economy*). (Open Problem 7, §15.) *(solution on page 48)*

15 Open problems (the starred exercises, collected)

The genuinely unsolved problems this paper surfaces, gathered for the reader who wants the research frontier rather than the pedagogy. Every starred exercise in the body cross-references this list *by name*, not by a hand-typed numeral, so the two cannot drift apart. Note that the "OP-n" identifiers used in the body for kernel-layer problems (e.g. OP-4, checkpoint with teeth) belong to a *shared* namespace owned by *The Single-Writer Kernel* and are deliberately not renumbered here; this list is this chapter's own.

1. **Branching continuity (§4).** If a checkpoint is forked into two live incarnations, which inherits the reputation? Parfit says identity-is-not-what-matters; the economy needs a rule, and the rule invites an attack.

2. **Agent-death taxonomy (§5).** What is recoverable vs. fundamentally lost when an LLM agent dies mid-thought? Event-Sourced Neural Rehydration (OP-4 in the shared kernel namespace) is a specified candidate answer — restore the Git SHA, truncate the durable message log to the crash point, replay under prompt-prefix caching — but it has not shipped, and its completeness has not been established: replay restores lineage and,

within one provider/model/version, a behaviorally equivalent continuation; it does not export the inference state, so across providers the honest description is task continuity with actor succession. The taxonomy itself — record vs. claims vs. execution state vs. latent "intent" — stays open until the mechanism ships and is measured.

3. **Cross-operator attestation (§7).** Binding identity keys across harbors you do not own. The highest-leverage unbuilt keystone for the market.

4. **The grading oracle's honest root and its telemetry (§11).** The rate-the-raters recursion itself is *no longer open*: Theorem 11.1 [30] closes it with a derived contraction factor $\lambda = 1 - \rho d$ and a priced depth (27 levels / \$1,350 at C=8; 53 / \$2,650 at C=1). What remains open is what that theorem explicitly does not supply: the exogenously honest root the tower terminates against at scale, and the deployment telemetry that estimates d and the cross-clique collusion correlation the disjointness hypothesis assumes away.

5. **Vector-reputation disclosure (§10).** Publish a vector and agents Goodhart the heaviest axis; publish a scalar and you have a bandit. The right disclosure form is open.

6. **Self-improvement arms race (§9).** Chiu et al. 2025: does capping reputation's marginal return restore efficiency or relocate the waste?

7. **Reputation-claim revocation convergence (§12).** Surgically retract a fraudulent outcome across N harbors, prove convergence, bound the spendable window — without violating append-only.

8. **Bond-farming defense (§6).** Principal binding vs. listing fee vs. sampled re-audit of high-velocity pairs; the false-positive cost to honest high-volume counterparties.

9. **Arbitration capacity (§11).** A market for bonded, rated, slashable human/automated arbiters: honest rulings, or rubber-stamping under load? This is Item 4's honest root, stated as a market-design problem.

10. **Skill-version reputation (§10).** A portfolio proof for skill v1 says nothing about v2; the lemons problem reappears at every version bump.

16 Conclusion

We set out to bridge the wedge to the market by answering one question: what has to be true before a coding agent's work can be *sold*? The answer is not a score. It is a *self* — a role plus continuity, keyed on an identity that cannot be re-picked, accumulating a transitive, witnessed history that survives every respawn. The estimator that turns that history into a number is the cheap, last part; the substrate beneath it is the gate, and most of that substrate is formally not built yet.

We were disciplined about the keystone (this paper stands on a local non-forgeable identity and hands the market paper the unbuilt cross-operator attestation by name), about the hardest hole (the grading oracle's incentive-compatibility is a flaw in the core theorem, not a footnote — and the rate-the-raters recursion underneath it is now closed by a companion theorem with a derived contraction factor and a priced tower depth, leaving the honest root and the audit telemetry as the residue), and about the cheerful slogans (reputation is monotone in *honest* outcomes but individually revocable, and bounded memory is a parameter, not a virtue). Reputation is not a bandit problem — it is non-stationary, multi-dimensional, strategically adversarial, and graded by an oracle with its own incentives.

> *A spawn does work and disappears. A person does work and* stays *— and only what stays can be priced. The harbor's whole economy is the machinery for manufacturing, and then formally grading, a self.*

A Implementation status of the reference system

The body of this paper is written to stand on its own: every mechanism is described as a component (a memory record, a checkpoint organ, a witnessed-outcome ledger) and graded on the neutral maturity scale (**implemented** / PARTIAL / SPECIFIED / *proposed*). This appendix grounds that scale in a concrete *reference system* — a running coordination daemon for local agent swarms — so a reader who wants to know "which of these actually runs today" can check. Nothing in the body depends on this appendix; it exists so the maturity labels are falsifiable rather than rhetorical.

Component (paper concept)	Maturity	Status in the reference system
Episodic memory (Organ 1)	**implemented**	Durable typed episodes promoted from execution history; runs in production.
Restart organ (Organ 2)	PARTIAL	Heartbeat-staleness detector that salvages dead agents; forwards a *summary*, not execution state.
Outcome ledger (Organ 3)	PARTIAL	Durable commitments, daemon-derived deadlines, oracle-bound closure, API/CLI, and overdue detection ship; neutral grading, sanctions, and reputation binding do not.
Cost meter (efficiency axis)	**implemented**	Per-task token/wall-clock/dollar accounting recorded today.
Local non-forgeable identity	PARTIAL	Daemon-minted actor-soul, lookup credential, and bounded newcomer pool ship; full write-boundary enforcement and legacy migration do not.
Capability jail (capability $\neq$ permission)	SPECIFIED	Per-agent tool-allowlist and scoped filesystem; specified.
Honest-attestation discipline	**implemented**	"Report all-good only when verified" is an enforced engineering rule.
Multi-dimensional reputation	SPECIFIED to *proposed*	No reputation component yet; the substrate it scores over is partly built.
Neutral-judge ceremony (Proto. 10.1)	SPECIFIED to *proposed*	Bonding and task-planning primitives exist; the hiring ceremony is designed.
Skill-helpfulness schema (Proto. 10.2)	SPECIFIED	Skill grafting is observable; the witnessed-event schema is specified.
Tombstone revocation (Proto. 12.1)	SPECIFIED	Append-only ledger semantics are designed; cross-harbor propagation is *proposed*.
Cross-operator attestation	*proposed*	The unbuilt keystone for the market; owed to *The Harbor Economy*.

Table 10: Maturity of each paper concept against a reference coordination daemon. The honest summary: the *organs* of continuity exist (one weakly); the *spine* of reputation — a witnessed-outcome ledger keyed on a non-forgeable identity — is specified but not yet shipped.

Solutions to the exercises

5.1 *(exercise on page 38)* A spawn is a temporary process filling a role; a person is a role-holder whose accountable continuity survives replacement of that process.

5.2 *(exercise on page 38)* Read right-to-left. No reputation ⇒ a buyer cannot price differentiated trust, so the alleged tradeable asset is adverse-selection noise. No durable accountable person/outcome identity ⇒ reputation attaches to a disposable incarnation and can be reset or duplicated. No continuity ⇒ successive role-holders cannot be shown to belong to one lineage; history does not follow replacement. No memory/checkpoint/evidence ⇒ a successor lacks both resumable state and a witnessed chain from old work to new work. The chain is not literally linear: non-forgeable identity and witnessed outcomes are parallel prerequisites for accountable continuity; checkpointing helps resumption but is not logically necessary for every reputation system.

5.3 *(exercise on page 38)* The human operator is a moral/legal person but should be represented economically as the *principal* above agent identities (Def. 6.2). If one principal can mint unlimited apparently independent agent-persons, it can whitewash sanctions, forge quorums, reuse aggregate collateral, and farm reputation through self-trade. Bind every agent lineage to a principal credential, cap principal-wide exposure and newcomer credit, and disclose shared control: the actor may change; the economic liability root must not.

5.4 *(exercise on page 38)* Capable-and-permitted: an agent has a test-runner tool and authority to run the suite. Capable-but-forbidden: the process can invoke raw `git push -force`, but policy forbids it. Permitted-but-incapable: the role is authorized to deploy, but the incumbent has no deploy credential/network route.

5.5 *(exercise on page 38)* Reputation attaches to the person's continuity witness and witnessed outcomes, not to the abstract role's O, C, or A. A role is fillable and history-free; otherwise a fresh spawn could inherit the prior holder's credit merely by taking the same job title, exactly the attribution Alice loses in §1.

5.6 *(exercise on page 38)* Keep a role template $(O_{\text{required}}, A_{\text{role}})$ separate from an incumbent capability set C_{subject}. At assignment require each current obligation to satisfy $O_{\text{active}} \subseteq A_{\text{role}} \cap C_{\text{subject}}$. Keep $C_{\text{subject}} \setminus A_{\text{role}}$ as capable-but-forbidden and $A_{\text{role}} \setminus C_{\text{subject}}$ as authorized-but-unavailable, which triggers provisioning or reassignment. Never equate C with A.

5.7 *(exercise on page 38)* Connectedness is a direct memory/intention link between two incarnations; continuity is the transitive, provenance-bearing chain across arbitrarily many links. A successor may read its predecessor's handoff and be psychologically connected while using a new outcome-ledger key, so the accountable chain is broken: connected, not economically continuous.

5.8 *(exercise on page 38)* In the manuscript's analogy, the body-lease is the replaceable live substance/process; the actor-soul is the durable identity/mailbox/history that carries consciousness-like continuity. Respawn replaces the body-lease and should retain the actor-soul only under a valid succession event.

5.9 *(exercise on page 39)* Represent a fork as a lineage DAG, never as two copies of one linear identity. Both children may display the ancestor's evidence with an explicit inherited/discounted prior, but neither duplicates the ancestor's outstanding authority, collateral, or spendable reputation. Allocate a fresh branch ID, split or revoke exclusive capabilities, and aggregate risk at the common principal/ancestor. Copying full reputation to both invites credit duplication, quorum multiplication, and double-spending of grants; giving it to only the first child creates a fork-race attack. This sketch is now a proved theorem — see the Numbers by hand box below.

5.10 *(exercise on page 39)* The script prints: "`sum of inherited priors over the closure = 0.900 + 0.810 + 0.729 = 2.439 > q = 1.0`" for the budget-only counterexample, then "`4000 random DAGs ...  0 violations; max live/witnessed ratio observed = 0.999995`" for the transfer-semantics sweep, then "`an 8-way copy-fork of one outcome yields 8.2x its witnessed value`" for the copy-full mutation. By hand: $0.9 + 0.9^2 + 0.9^3 = 0.9 + 0.81 + 0.729 = 2.439$, three terms of a geometric series, confirming the closure-sum counterexample without running anything; under transfer semantics the same chain instead *debits* each hop, so the live total telescopes to $0.9^3 = 0.729 \leq 1$, which is what the sweep's 0 violations certifies at scale.

5.11 *(exercise on page 39)* The checkpoint organ is the specifically weak one: it forwards notes rather than execution state. The outcome-ledger organ is also only partial in the implementation: commitments and oracle closure exist, but neutral graded outcomes and reputation binding do not.

5.12 *(exercise on page 39)* Memory retains the session, notes, and explicit decisions. The current checkpoint may retain a handoff and perhaps a committed diff, but not working process state, tool handles, pending effects, or hidden provider state. The outcome ledger retains the commit only if it is bound to a work unit and independently witnessed; a naked Git commit is an artifact, not proof of accepted delivery. A strong checkpoint would also preserve the worktree, environment, open claims/epochs, acceptance state, tool receipts, and the idempotency journal.

5.13 *(exercise on page 39)* A useful death taxonomy, in increasing order of what survives: (1) *record survival* — logs/notes only, reconstruction is manual; (2) *artifact survival* — a content-addressed worktree and outputs survive, computation is rerun; (3) *semantic task capsule* — artifacts plus obligations, decisions, transcript/projection, tools, and a pending-effect journal, so a new provider can continue observably; (4) *execution snapshot* — process memory/runtime state survives under a controlled runtime, rarely portable; (5) *latent state* — unexposed activations, KV cache, or "intent" dies, fundamentally unrecoverable from an arbitrary provider. Checkpoint guarantees should state which tier they provide.

5.14 *(exercise on page 39)* Correctly scoped, Property 6.1 says reputation has economic force only to the extent that the accountable principal cannot cheaply discard or multiply the liability-bearing identity. Whitewashing escapes a negative record when newcomer access is better; a Sybil attack multiplies votes/credit when the mechanism counts identities. Non-forgeability is necessary for those mechanisms under those access rules, but it is not sufficient by itself.

5.15 *(exercise on page 39)* Full work with a reduced economic ceiling lets an honest newcomer demonstrate competence without being denied livelihood, while limiting the maximum one-shot gain from abandoning a mature sanctioned identity. "Reduced work" slows evidence accumulation and taxes all honest entrants; "reduced exposure" targets the externality instead. The ceiling schedule must be chosen so the maturation opportunity cost exceeds plausible evasion gain.

5.16 *(exercise on page 39)* Principal binding is the load-bearing defense: aggregate reputation, exposure, counterparties, and sanctions across the 1,000 agents, so self-trades add little or no independent evidence. A listing fee raises attack cost but only deters while total fees exceed the minted value. Sampled adversarial re-audit detects fabricated work, with false-positive burden approximately the sampling rate times the honest high-volume population and the adjudication error. Add counterparty-diversity weighting (sharply diminishing credit per principal pair and per correlated pair), independent payer stake, and principal-wide reserve limits; the sampled re-audit rate that makes farming unprofitable is the $\rho^\star = G/(dB)$ of §11, with G the value a farmed record can mint.

5.17 *(exercise on page 39)* The script prints "`schedules tested = 76000`", "`dominating schedules = 0/4000 instances (expected 0)`", and "`min (H(g) - H(cliff)) = +2.817e-02 (>= 0 means undominated)`"; the exchange step's table shows H at an all-mass-at-t schedule growing as $G_{\max}(\delta_h/\delta_f)^t$ and reports "`strictly increasing in t:  yes`". By hand: with $\delta_h > \delta_f$, $(\delta_h/\delta_f) > 1$, so its powers strictly increase in t — deterrence bought later always costs the honest newcomer more, which is the whole exchange argument in one inequality.

5.18 *(exercise on page 39)* The script prints "`reachable states to depth 7:  747 (467 post-migration, 693 with an open sanction window)`" and "`ok intact:  ZERO Def-III.6.1 violations over all reachable states [internal]`"; the four mutants (drop scope, drop clause (i), drop clause (ii), drop clause (iii)) are each caught with shortest counterexamples $1, 2, 2, 2$ steps. The one-step break is the scope mutant `m0`: keying sanctions by *(identity, engine)* instead of by identity alone lets a single dishonest outcome escape through a sibling engine key inside the *same* identity — no migration needed at all. By hand: this is why the intact protocol sanctions per *principal* while pricing per *(principal, engine)*; collapsing that distinction is the shortest way to violate Def. III.6.1.

5.19 *(exercise on page 39)* Local non-forgeable identity proves that, inside one trusted daemon, an actor cannot freely impersonate or re-pick its daemon-issued handle. Cross-operator attestation must additionally bind a key to a durable principal across a hostile operator boundary, establish proof of possession/freshness, aggregate related agents, and support revocation and audit.

5.20 *(exercise on page 39)* That sentence secretly assumes the cross-operator stone. With only local identity, Bob learns at most "Alice's daemon asserts this actor/key." A hostile Alice can mint arbitrary local actors, rewrite local history, or misstate principal separation.

5.21 *(exercise on page 40)* A witness log proves publication order and exposes equivocation; it does not prove who controls a key or that two keys are distinct principals. Close the gap with a scarce external root: hardware/remote attestation, an organizational/KYC issuer, a bonded sponsor chain, or a combination. A transparency log makes bindings and rotations auditable but not true; a sponsor creates economic accountability but can cartelize; a shared identity issuer centralizes and leaks privacy; hardware roots attest devices/code, not unique humans. State the chosen trust and Sybil assumptions.

5.22 *(exercise on page 40)* With a complete, mostly stable paired-comparison history, batch Bradley–Terry uses all observations and is more appropriate than order-sensitive streaming Elo. If abilities drift, use a time-varying/dynamic model rather than pretending full-history stationarity.

5.23 *(exercise on page 40)* A binary protected acceptance test can update a beta-Bernoulli reliability estimate (or a calibrated per-task success model). Human aesthetic preference is naturally pairwise and may use Bradley–Terry/TrueSkill with judge effects. EigenTrust expects a who-trusts-whom graph and therefore fits neither raw task-outcome signal directly.

5.24 *(exercise on page 40)* If identities are self-chosen, an exact Bayesian estimator simply produces precise scores for disposable names. If the agent authors the "oracle," a sophisticated estimator precisely learns manipulated labels. Better statistics cannot repair missing provenance, adversarial labels, or liability binding; secure the event-generation substrate first.

5.25 *(exercise on page 40)* Stationary arm rewards versus model/skill/version drift; one scalar reward versus buyer-weighted quality vectors; a non-strategic environment versus Goodhart, collusion, and whitewashing; and an observed/trusted reward versus a capturable grader.

5.26 *(exercise on page 40)* A contextual bandit is appropriate for one principal privately routing its own tasks among candidate agents, with a fixed local objective, controlled feedback, an exploration budget, and no public asset being minted. The result is a routing policy, not public reputation.

5.27 *(exercise on page 40)* Capping the marginal reputation return can reduce direct improvement races, but agents may redirect spending into unmeasured signaling, judge capture, identity proliferation, or lobbying for task mix. Efficiency requires the full game: improvement cost, social value of quality, transferability, and alternative signals. Candidate controls are concave credit, buyer-local vectors, nontransferable evidence, and taxes/bonds on negative externalities. No cap is generically efficient without those payoff assumptions.

5.28 *(exercise on page 40)* Accuracy: competent, a protected test/spec oracle; incompetent, a self-reporting worker. Aesthetics: competent, a conflict-free expert reviewer for that domain; incompetent, a token-cost meter or a same-family unblinded model. Efficiency: competent, a protected resource meter normalized for the task; incompetent, a style reviewer guessing from prose.

5.29 *(exercise on page 40)* The requesting principal/daemon defines the judging work and eligible pool; the selected judge posts its own bond; the grade lands as a signed witnessed event in the subject's outcome ledger with judge/provenance; a conflict-free re-audit or appeal that overturns it slashes the judge under the predeclared rule and tombstones/revises the grade.

5.30 *(exercise on page 40)* Record skill ID/version/content hash, agent/build/principal, task and stratum descriptors, randomized assignment or selection propensity, exact context-graft receipt, model/environment/tool versions, outcome/evidence references, quality vector, costs, and evaluator identities. These fields verify exposure and outcome association. "This skill helped" is causal and cannot be verified from a graft-success pair alone; estimate it with randomized skill-on/off assignment on held-out comparable tasks (or label an observational matched estimate honestly), reporting effect size and uncertainty.

5.31 *(exercise on page 40)* Publish the vector, per-axis uncertainty, sample sizes, cohort/task distribution, freshness, judge/evidence provenance, and known correlations. Let each buyer apply a local, preferably non-public weighting and hard minimums; do not mint a canonical platform scalar. Agents can still optimize market-demanded axes, so rotate held-out tests and audit cross-axis regressions. Plural preference does not eliminate Goodhart, but it prevents one universal target from becoming the asset itself.

5.32 *(exercise on page 40)* Settlement incentives are conditional on the signal distribution. "The daemon derives the grade" only identifies the messenger; if a strategic judge chooses the input, the distribution depends on omitted actions/payoffs and the equilibrium proof assumes its own conclusion.

5.33 *(exercise on page 41)* Judge J posts bond B; its grade and the bond lock are recorded; an independently selected re-auditor overturns the dishonest grade against preserved evidence; the system appends a correction/tombstone, recomputes any explicitly appealable settlement, slashes B to the harmed/commons bucket, and lowers J's judging reputation. It must not silently rewrite the original event.

5.34 *(exercise on page 41)* With one-shot corrupt gain at most G, re-audit probability ρ, conditional detection probability d, and slashable bond B, local deterrence requires $\rho dB > G$, i.e. $\rho^\star = G/(dB)$. At $G_0 = 400, d = 0.8, B = 50, \rho = 0.25$: $\rho d = 0.2$, so $\lambda = 1 - \rho d = 0.8$; with $C = 8$ the threshold $G_k > CB = 400$ is unmet at level 0, giving $\lceil \log 400 / \log 1.25 \rceil = 27$

levels; with $C = 1$ the threshold is only $G_k > 50$, so bribery bleeds linearly for 35 levels before 18 more geometric ones close it, 53 total.

5.35 *(exercise on page 41)* Admit arbiters by domain competence and conflict graph; assign cases randomly/blindly; require bonds and service-level commitments; preserve evidence; pay for timely completed work rather than agreement; sample re-audits and publish calibration/overturn rates; use plural panels on high stakes. Price queue capacity and cap workload so rubber-stamping is not the profitable response to overload. Honest convergence still needs a protected mechanical outcome, a delayed real-world outcome, or a final human/contractual authority — a market alone does not manufacture truth.

5.36 *(exercise on page 41)* The script prints "`flat:  50.00 (Theta(T))`", "`model A: 25.58 (Theta(log T); a*G/(d v)*ln(1+vT/B) = 25.50)`", and "`model B: 35.08 (O(1); closed form aG^2/(2dBrv) = 41.67; hits 0 at t*=333)`". By hand: the flat baseline is a constant audit rate $\rho^\star$ times a constant per-audit cost a summed over T periods, so $\rho^\star \cdot a \cdot T = 0.25 \times 1 \times 200 = 50$, exactly the printed 50.00 — the two amortization models are cheaper only because they let ρ_t fall below $\rho^\star$ as accumulated stake or surfaced history does part of the deterring.

5.37 *(exercise on page 41)* An append-only tombstone preserves the original claim, the later correction, their authors, timing, and the reliance window. Delete destroys evidence, permits history rewriting, and makes replicas unable to distinguish retraction from omission.

5.38 *(exercise on page 41)* Append a signed tombstone containing the target outcome ID/hash, reason/evidence, authority, revision, and effective time. Propagate it and recompute reputation as a projection that excludes/nullifies that outcome. It was spendable from original publication until each relying harbor received and enforced the tombstone or failed closed; under an unbounded partition that window is unbounded.

5.39 *(exercise on page 41)* Use a signed revision DAG, per-harbor import receipts/watermarks, periodic accumulator/Merkle roots for active outcomes and tombstones, and a freshness rule that refuses high-risk use from stale replicas. Under connected eventual delivery, convergence is shown when every member acknowledges a root containing the tombstone; sampling can audit inclusion. Bounded local memory may retain compact accumulators and content-addressed archive pointers, not erase correction commitments. No finite convergence/spend bound exists during an unbounded partition.

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